

An explicit exponential pair problem and sharp rotational conformal-Killing pinching

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Abstract

We determine the minimum of an exponential interaction over continuous functions with prescribed lower and upper obstacles. The minimizer consists of two constant pieces joined by an explicit free arc. Its value is given for every cap, and an exact convex deficit identity proves uniqueness. At every nondegenerate cap the minimizer has derivative jumps at both contacts; every admissible function differentiable in the interior has strictly larger value. The deficit also controls the squared L^2 distance to the minimizer with an explicit positive cap-dependent constant. Applying this estimate to two hemispheres gives quantitative reflection symmetry without imposing symmetry as a constraint. For smooth sphere metrics obtained from positive balanced rotational profiles and their diffeomorphic pullbacks, a complete Haar decomposition gives nonnegativity and zero-remainder rigidity without pinching. The variational solution determines the exact conformal-Killing curvature threshold, with smooth sharpness and full equality classification at the safe threshold. A local plateau variation, strict curvature contraction and exact-moment smooth recovery give a genuine negative one-form below that threshold. Consequently the unrestricted rotational threshold is strictly smaller.

1 The constrained problem

For $C \geq 1$, consider real functions whose restrictions to the unit interval are continuous, and let

$$\mathcal{A}_C = \{k : \mathbb{R} \rightarrow \mathbb{R} : k|_{[0,1]} \in C([0,1]), \quad 0 \leq k(v) \leq \log C \text{ for } v \in [0,1]\},$$

$$\mathcal{I}(k) = \int_0^1 \int_0^s 2se^{k(v)-k(s)} dv ds.$$

All functions and integrals in this paper are real-valued. The interval constraints include both endpoints. In particular $\mathcal{A}_1$ contains only functions vanishing on $[0,1]$. Values outside this interval do not affect the functional; uniqueness always means equality on $[0,1]$.

Theorem 1.1. *For every $C \geq 1$ there is a unique $q \geq 1$ satisfying*

$$\log q + \frac{2}{3}(q^3 - 1) = \log C. \tag{1}$$

Set

$$\alpha = \frac{1}{\sqrt{q^4 + 2q}}, \quad \beta = \alpha q^2. \tag{2}$$

Then $0 < \alpha \leq \beta < 1$, and the function $k_* = \log E_*$, where

$$E_*(v) = \begin{cases} 1, & v \leq \alpha, \\ \sqrt{v/\alpha} \exp\left(\frac{2}{3}((v/\alpha)^{3/2} - 1)\right), & \alpha \leq v \leq \beta, \\ C, & v \geq \beta, \end{cases} \quad (3)$$

belongs to $\mathcal{A}_C$. For every $k \in \mathcal{A}_C$,

$$\mathcal{I}(k) \geq \frac{2}{3} - \beta + \frac{1}{3\beta}, \quad (4)$$

with equality if and only if $k = k_*$ on $[0, 1]$. At $C = 1$ the profile is identically zero and the minimum is $2/3$.

The degeneracy in the middle interval is essential at cap one. No derivative formula on a nonempty free interval is needed for that endpoint.

Lemma 1.2. *The parameter in (1) exists uniquely and increases strictly with $C \geq 1$. It equals one precisely when $C = 1$. The contacts satisfy*

$$\beta^2 = \frac{q^3}{q^3 + 2}, \quad 1 - \beta^2 = 2\alpha^2 q, \quad \frac{\alpha}{q} = \frac{1 - \beta^2}{2\beta}. \quad (5)$$

The contacts are distinct precisely when $C > 1$.

Proof. The function $F(q) = \log q + \frac{2}{3}(q^3 - 1)$ is continuous and strictly increasing on $[1, \infty)$. It has $F(1) = 0$ and $F(C) \geq \log C$ for $C \geq 1$, so the intermediate value theorem on $[1, C]$ gives the unique parameter. Strict increase of F and of the logarithm gives the claimed dependence on C . Squaring (2) gives (5). In particular $\beta < 1$, while $q \geq 1$ gives $\alpha \leq \beta$, strictly when $q > 1$. $\square$

The free expression in (3) equals one at $v = \alpha$ and $q \exp(\frac{2}{3}(q^3 - 1)) = C$ at $v = \beta$. Both its square-root and exponential factors are nonnegative and increasing for positive v . It therefore joins the obstacles continuously and stays between their values. This proves the asserted admissibility.

2 Exact primitives and marginal

Write $\phi(v) = \frac{2}{3}((v/\alpha)^{3/2} - 1)$ on the free arc, and define the actual prefix and tail integrals

$$P(v) = \int_0^v E_*(t) dt, \quad Q(v) = \int_v^1 \frac{2s}{E_*(s)} ds.$$

Lemma 2.1. *The two integrals have the formulas*

$$P(v) = \begin{cases} v, & v \leq \alpha, \\ \alpha e^{\phi(v)}, & \alpha \leq v \leq \beta, \\ C(v - \beta + \alpha/q), & v \geq \beta, \end{cases} \quad Q(v) = \begin{cases} 3\alpha^2 - v^2, & v \leq \alpha, \\ 2\alpha^2 e^{-\phi(v)}, & \alpha \leq v \leq \beta, \\ (1 - v^2)/C, & v \geq \beta, \end{cases} \quad (6)$$

for $v \in [0, 1]$. Consequently the marginal density $A(v) = E_*(v)Q(v) - 2vP(v)/E_*(v)$ is

$$A(v) = \begin{cases} 3(\alpha^2 - v^2), & 0 \leq v \leq \alpha, \\ 0, & \alpha \leq v \leq \beta, \\ -3(v - \beta)(v + 1/(3\beta)), & \beta \leq v \leq 1. \end{cases} \quad (7)$$

Proof. Direct differentiation on the positive half-line gives

$$\frac{d}{dv}(\alpha e^{\phi(v)}) = E_*(v), \quad \frac{d}{dv}(-2\alpha^2 e^{-\phi(v)}) = \frac{2v}{E_*(v)}$$

on the free arc. At its upper endpoint the first primitive has value $C\alpha/q$, while the second has value $-(1 - \beta^2)/C$ by (5). Splitting the integrals at the contacts gives (6); the calculation also applies when the contacts coincide. Substitution into the definition of A proves the first two pieces of (7). On the upper piece it gives $1 - 3v^2 + 2v\beta - 2v\alpha/q$, which has the stated factorization by the last identity in (5). $\square$

3 Convex calibration and rigidity

For continuous k, g on $[0, 1]$, put $\delta = k - g$ and

$$A_g(v) = e^{g(v)} \int_v^1 2se^{-g(s)} ds - 2ve^{-g(v)} \int_0^v e^{g(t)} dt.$$

Lemma 3.1. *The exact identity*

$$\begin{aligned} \mathcal{I}(k) - \mathcal{I}(g) &= \int_0^1 A_g(v) \delta(v) dv + \mathcal{R}(k, g), \\ \mathcal{R}(k, g) &= \int_0^1 \int_0^s 2se^{g(v)-g(s)} \left[e^{\delta(v)-\delta(s)} - 1 - \delta(v) + \delta(s) \right] dv ds \end{aligned} \tag{8}$$

holds. The remainder is nonnegative, and it vanishes if and only if δ is constant on $[0, 1]$.

Proof. Apply the identity $e^{x+y} - e^x = e^x y + e^x(e^y - 1 - y)$ with $x = g(v) - g(s)$ and $y = \delta(v) - \delta(s)$. Fubini's theorem identifies the linear term with the stated marginal integral. All integrands are continuous on the compact square before restriction to the triangle, so these integrations are justified. The inequality $e^y \geq 1 + y$ proves nonnegativity. If the remainder vanishes, its nonnegative integrand is zero almost everywhere. Continuity makes it zero at every point of the open triangle $0 < v < s < 1$: otherwise it would be positive on an open set of positive measure. The prefactor is positive there, and strictness of $e^y > 1 + y$ for $y \neq 0$ gives $\delta(v) = \delta(s)$. Any two interior points can be ordered, so δ is constant in the interior, and continuity extends that constant to the endpoints. The converse follows directly from the integrand. $\square$

Proof of Theorem 1.1. Take $g = k_*$. On the lower obstacle $\delta = k \geq 0$ and $A \geq 0$; on the upper obstacle $\delta = k - \log C \leq 0$ and $A \leq 0$; on the free arc $A = 0$. Thus both terms in (8) are nonnegative, and k_* is a minimizer. If equality holds, Lemma 3.1 gives $k - k_* = c$ on the interval. At zero the lower constraint gives $c \geq 0$, while at one the upper constraint gives $c \leq 0$. Therefore $c = 0$.

It remains to evaluate the value. Fubini gives $\mathcal{I}(k_*) = \int_0^1 2vP(v)/E_*(v) dv$. The integrand on the three pieces is, respectively,

$$2v^2, \quad 2\alpha^2 \sqrt{v/\alpha}, \quad 2v(v - \beta + \alpha/q).$$

Its integral is consequently

$$\frac{2}{3}\alpha^3 + \frac{4}{3}\alpha^3(q^3 - 1) + \frac{2}{3}(1 - \beta^3) - (\beta - \alpha/q)(1 - \beta^2).$$

The identities (5) reduce this expression to $\frac{2}{3} - \beta + 1/(3\beta)$. The integration of the middle term is valid also on a degenerate interval. Alternatively, at $C = 1$ one directly has $\mathcal{I}(0) = \int_0^1 2s^2 ds = 2/3$. $\square$

Theorem 3.2. *For $C > 1$, the one-sided derivatives of the minimizer are*

$$k'_{*,-}(\alpha) = 0, \quad k'_{*,+}(\alpha) = \frac{3}{2\alpha}, \quad k'_{*,-}(\beta) = \frac{1}{2\beta} + \frac{q}{\alpha}, \quad k'_{*,+}(\beta) = 0.$$

In particular, every $k \in \mathcal{A}_C$ differentiable on $(0, 1)$ satisfies strict inequality in (4).

Proof. The obstacles are constant. In the free arc direct differentiation gives $k'_*(v) = 1/(2v) + \sqrt{v/\alpha}/\alpha$. Since $\alpha < \beta$, its one-sided limits give the displayed positive derivatives. Both contacts are interior, so the jumps exclude differentiability there. Equality in (4) would identify a differentiable competitor with k_* throughout the interval, contradicting either jump. $\square$

4 A quantitative exponential estimate

Lemma 4.1. *For real m, x, y with $m \leq x, y$,*

$$e^x - e^y - e^y(x - y) \geq \frac{e^m}{2}(x - y)^2.$$

Proof. The function $t \mapsto e^t - e^m t^2/2$ has nonnegative second derivative on $[m, \infty)$. Its supporting-line inequality at y , applied to x , is exactly the stated bound after rearrangement. This applies in either order of x and y , including the boundary value m . $\square$

5 Quantitative stability

For a continuous real function δ on $[0, 1]$, define

$$\mathcal{Q}(\delta) = \int_0^1 \int_0^s 2s(\delta(v) - \delta(s))^2 dv ds, \quad \mu = \int_0^1 \delta, \quad g = \delta - \mu, \quad G(x) = \int_0^x g.$$

Lemma 5.1. *The actual weighted pair integral satisfies*

$$\mathcal{Q}(\delta) = \int_0^1 (1 + x^2)g(x)^2 dx + 2 \int_0^1 G(x)^2 dx. \quad (9)$$

In particular $\mathcal{Q}(\delta) \geq \int_0^1 g^2$.

Proof. Centering does not change pair differences, and $\int_0^1 g = 0$. Let $H(x) = \int_0^x g(v)^2 dv$. Expanding the inner integral gives

$$\mathcal{Q}(g) = \int_0^1 (2xH(x) - 4xg(x)G(x) + 2x^2g(x)^2) dx.$$

The function $F(x) = (x^2 - 1)H(x) - 2xG(x)^2$ is continuous up to the endpoints and differentiable in the interior. Direct differentiation gives

$$F'(x) = 2xH(x) - 4xg(x)G(x) + 2x^2g(x)^2 - ((1 + x^2)g(x)^2 + 2G(x)^2).$$

Since $G(0) = G(1) = H(0) = 0$, both endpoint values of F vanish. The fundamental theorem of calculus proves (9). All integrands are continuous; no derivative of δ is required. The final inequality follows from nonnegativity of the additional terms. $\square$

Theorem 5.2. For $C \geq 1$, let α and k_* be as in Theorem 1.1, and define

$$c(C) = \left(\frac{12C}{5\alpha} + \frac{\log C}{2\alpha^3} \right)^{-1} > 0. \quad (10)$$

Every $k \in \mathcal{A}_C$ satisfies

$$\mathcal{I}(k) - \mathcal{I}(k_*) \geq c(C) \int_0^1 |k(v) - k_*(v)|^2 dv. \quad (11)$$

Proof. At $C = 1$, both profiles vanish on the unit interval, so the assertion is immediate. For $C > 1$, set $L = \log C$, $\delta = k - k_*$ and $D = \mathcal{I}(k) - \mathcal{I}(k_*)$. Write the exact deficit as $D = \mathcal{P} + \mathcal{R}$, where $\mathcal{P} = \int_0^1 A(v)\delta(v) dv \geq 0$. For every pair $v < s$, the differences $x = k(v) - k(s)$ and $y = k_*(v) - k_*(s)$ are at least $-L$. Lemma 4.1 with $m = -L$ gives

$$e^y(e^{\delta(v)-\delta(s)} - 1 - \delta(v) + \delta(s)) \geq \frac{1}{2C}(\delta(v) - \delta(s))^2.$$

Multiplication by $2s$ and integration yield $\mathcal{R} \geq a\mathcal{Q}(\delta)$, where $a = 1/(2C) > 0$. With μ, g as above and $V = \int_0^1 g^2$, Lemma 5.1 therefore gives

$$\mathcal{P} + aV \leq D. \quad (12)$$

To control the constant mode, use the complete lower-obstacle weight

$$w(v) = \frac{3 \max\{\alpha^2 - v^2, 0\}}{2\alpha^3} \quad (0 \leq v \leq 1).$$

This is continuous and nonnegative. Polynomial integration gives

$$\int_0^1 w = 1, \quad \int_0^1 w^2 = \frac{6}{5\alpha}, \quad r := \int_0^1 (w - 1)^2 = \frac{6}{5\alpha} - 1 \geq 0. \quad (13)$$

On $[0, \alpha]$, the lower obstacle gives $\delta = k \in [0, L]$ and $A(v) = 3(\alpha^2 - v^2)$. Elsewhere $w = 0$ and $A\delta \geq 0$. Thus the pointwise bound $\delta^2 \leq L\delta$ on the support of w proves

$$\int_0^1 w\delta^2 \leq b\mathcal{P}, \quad b = \frac{L}{2\alpha^3} \geq 0. \quad (14)$$

Put $u = \int_0^1 w\delta$. Positivity of $\int_0^1 w(\delta - u)^2$ and $\int_0^1 w = 1$ imply $u^2 \leq \int_0^1 w\delta^2 \leq b\mathcal{P}$. The identity $\int_0^1 (w - 1)g = u - \mu$ and Cauchy–Schwarz give $(u - \mu)^2 \leq rV$. This instance of Cauchy–Schwarz follows directly by expanding the nonnegative quadratic integral $\int_0^1 (t(w - 1) + g)^2$ for every real t . Consequently

$$|\mu| \leq \sqrt{b\mathcal{P}} + \sqrt{rV}, \quad \mu^2 \leq (b + r/a)(\mathcal{P} + aV) \leq (b + r/a)D.$$

The middle inequality is the two-dimensional Cauchy–Schwarz inequality applied to $(\sqrt{b}, \sqrt{r/a})$ and $(\sqrt{\mathcal{P}}, \sqrt{aV})$. Finally $\int_0^1 \delta^2 = V + \mu^2$, and (12) implies $V \leq D/a$. Combining these facts with (13) gives

$$\int_0^1 \delta^2 \leq \left(b + \frac{1+r}{a} \right) D = \left(\frac{\log C}{2\alpha^3} + \frac{12C}{5\alpha} \right) D.$$

The coefficient is positive because $C, \alpha > 0$ and $\log C \geq 0$. This proves (11). No optimality of $c(C)$ is asserted. $\square$

6 Two hemispheres and near-equatorial symmetry

Corollary 6.1. *Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be continuous on $[-1, 1]$ and satisfy $0 \leq h \leq \log C$ there, where $C \geq 1$. Define*

$$k_+(v) = h(v), \quad k_-(v) = h(-v), \quad \mathcal{D} = \mathcal{I}(k_+) + \mathcal{I}(k_-) - 2\mathcal{I}(k_*).$$

Then

$$\int_{-1}^1 |h(v) - k_*(|v|)|^2 dv \leq \frac{\mathcal{D}}{c(C)}, \quad (15)$$

$$\int_{-1}^1 |h(v) - h(-v)|^2 dv \leq \frac{4\mathcal{D}}{c(C)}, \quad (16)$$

$$\int_{-1}^1 \left| h(v) - \frac{h(v) + h(-v)}{2} \right|^2 dv \leq \frac{\mathcal{D}}{c(C)}. \quad (17)$$

No reflection or balance condition is imposed on h .

Proof. Apply Theorem 5.2 to each hemisphere. Splitting the integral at zero and substituting $v \mapsto -v$ on the southern half identifies their summed squared errors with the left side of (15). At each northern point, the elementary inequality $(x - y)^2 \leq 2x^2 + 2y^2$, applied to the two errors against k_* , bounds the reflection error. That error has equal integrals on the two halves, giving the factor four in (16). The symmetrization error is pointwise one quarter of the squared reflection error, proving (17). At cap one all three errors vanish directly. $\square$

7 Smooth balanced profiles

Let $a : \mathbb{R} \rightarrow \mathbb{R}$ be globally smooth, positive on $[-1, 1]$, and balanced:

$$\int_{-1}^1 va(v) dv = 0. \quad (18)$$

Define the actual moment coordinate and warping function by

$$x(v) = \int_0^v a(\xi) d\xi, \quad f(v) = 2 \int_v^1 \xi a(\xi) d\xi. \quad (19)$$

Lemma 7.1. *The functions x and f are globally smooth, with $x' = a$ and $f' = -2va$. The coordinate x is strictly increasing on $[-1, 1]$. Moreover, $f(-1) = f(1) = 0$, $f > 0$ on $(-1, 1)$, and there exist globally smooth real functions b, c such that, on the whole real line,*

$$f(v) = (1 - v^2)b(v), \quad a(v)^2 - b(v)^2 = (1 - v^2)c(v). \quad (20)$$

Here $b > 0$ on $[-1, 1]$ and $b(\pm 1) = a(\pm 1)$. For a constant profile $a = r > 0$, one may take $b = r$ and $c = 0$.

Proof. The derivative formulas follow from the fundamental theorem of calculus; they also imply global smoothness. Positivity of a makes the integral of a between any two distinct points of $[-1, 1]$ positive, proving strict monotonicity of x . The northern endpoint value is immediate and the southern one is exactly the balance condition. For $0 \leq v < 1$, the integrand in the

northern expression for f is positive on $(v, 1)$. For $-1 < v < 0$, balance rewrites that expression as $f(v) = -2 \int_{-1}^v \xi a(\xi) d\xi > 0$.

For a globally smooth function p with $p(t) = 0$, the fundamental theorem gives the smooth factorization

$$p(v) = (v - t) \int_0^1 p'(t + s(v - t)) ds.$$

Applying it successively at the two distinct roots $-1, 1$ gives a globally smooth factor of $1 - v^2$. Apply this first to f to obtain b . Differentiating $f = (1 - v^2)b$ at each pole and using $f' = -2va$ yields $b(\pm 1) = a(\pm 1)$. Interior positivity follows from $f > 0$ and $1 - v^2 > 0$; the endpoint values give positivity at the poles. Thus $a^2 - b^2$ vanishes at both poles, and the same factorization gives c . Finally, for $a = r$ direct integration gives $f(v) = r(1 - v^2)$, which verifies the stated constant-profile factors. $\square$

Theorem 7.2. *Let a satisfy the hypotheses of Lemma 7.1, and let b, c be the factors supplied there. On the standard unit sphere $S^2 \subset \mathbb{R}^3$, let z be the third coordinate and g_0 the round metric. Then*

$$g = b(z)g_0 + \frac{c(z)}{b(z)} dz \otimes dz \quad (21)$$

is a smooth positive Riemannian metric on the whole sphere, including both poles. For the constant profile $a = r > 0$ with $b = r, c = 0$, it is exactly rg_0 .

Proof. The height takes values in $[-1, 1]$, where b is positive, so both scalar coefficients in (21) are smooth on the sphere. The one-form dz is the actual differential of the smooth coordinate function, hence the displayed tensor is smooth at the poles as well as elsewhere. For $p \in S^2$ and $v \in T_p S^2$, orthogonality $p \cdot v = 0$ gives

$$dz_p(v) = (e_3 - z(p)p) \cdot v, \quad |dz_p(v)|^2 \leq (1 - z(p)^2)g_0(v, v).$$

Write $q = c(z)/b(z)$. If $q \geq 0$, positivity follows from $b(z) > 0$. If $q < 0$, the same bound and the second factorization in (20) give

$$g(v, v) \geq [b(z) + q(1 - z^2)]g_0(v, v) = \frac{a(z)^2}{b(z)}g_0(v, v) > 0 \quad (v \neq 0).$$

The constant-profile formula follows directly from $b = r, c = 0$. $\square$

8 Cylinder coordinates and area

For $-1 < v < 1$ and $\theta \in \mathbb{R}$, put $\rho(v) = \sqrt{1 - v^2}$ and

$$\Phi(v, \theta) = (\rho(v) \cos \theta, \rho(v) \sin \theta, v) \in S^2.$$

The angle is a real parameter, so no global injectivity is asserted.

Lemma 8.1. *The map Φ is smooth and has injective differential. Its height is v , and the metric of Theorem 7.2 satisfies*

$$\Phi^*g = \frac{a(v)^2}{f(v)} dv^2 + f(v) d\theta^2. \quad (22)$$

Proof. The identity $\rho^2 + v^2 = 1$ gives the sphere constraint, and $\rho > 0$ gives smoothness on the stated domain. The actual derivatives are

$$\Phi_v = \left(-\frac{v}{\rho} \cos \theta, -\frac{v}{\rho} \sin \theta, 1 \right), \quad \Phi_\theta = (-\rho \sin \theta, \rho \cos \theta, 0).$$

Their round inner products are ρ^{-2} , 0, and ρ^2 , respectively. In particular the differential is injective. Since $dz(\Phi_v) = 1$ and $dz(\Phi_\theta) = 0$, the radial coefficient of the pullback of (21) is

$$\frac{b}{1-v^2} + \frac{c}{b} = \frac{b^2 + (1-v^2)c}{(1-v^2)b} = \frac{a^2}{f}.$$

The angular coefficient is $(1-v^2)b = f$, and the mixed coefficient vanishes. These equalities give (22). $\square$

Theorem 8.2. *Write $d\mu_0$ for round area measure. The metric of Theorem 7.2 has the global area identity*

$$d\mu_g = a(z) d\mu_0. \quad (23)$$

For every $F : \mathbb{R} \rightarrow \mathbb{R}$ continuous on $[-1, 1]$,

$$\int_{S^2} F(z) d\mu_g = 2\pi \int_{-1}^1 a(v) F(v) dv. \quad (24)$$

Proof. The round gradient of height at p is $e_3 - z(p)p$, of squared length $1 - z(p)^2$. In a round orthonormal tangent basis, the determinant of $bg_0 + q dz \otimes dz$ is

$$b^2 + bq |dz|_{g_0}^2 = b^2 + c(1 - z^2) = a^2, \quad q = c/b.$$

Thus its area density relative to g_0 is $a > 0$. The computation is valid at the poles and in every chart, proving the measure identity (23).

For completeness, the round height disintegration follows directly from the two hemisphere graphs. On the open unit disk in $\mathbb{R}^2$, the upper graph $y \mapsto (y, \sqrt{1 - |y|^2})$ has area density $(1 - |y|^2)^{-1/2}$. The lower graph has the same density and negative height. The equator has round area zero. Radial integration and the substitution $u = \sqrt{1 - r^2}$ therefore give, for continuous G on $[-1, 1]$,

$$\int_{S^2} G(z) d\mu_0 = 2\pi \int_0^1 \frac{r [G(\sqrt{1 - r^2}) + G(-\sqrt{1 - r^2})]}{\sqrt{1 - r^2}} dr = 2\pi \int_{-1}^1 G(u) du.$$

All integrals converge: G is bounded and $r/\sqrt{1 - r^2}$ is integrable on $(0, 1)$. Applying this formula to $G = aF$ and using (23) proves (24). $\square$

Theorem 8.3. *The Gauss curvature of the metric in Theorem 7.2 is*

$$K_g(p) = \frac{1}{a(z(p))} \quad (p \in S^2). \quad (25)$$

Equivalently, its scalar curvature is $2/a(z)$, and every nondegenerate tangent two-plane has sectional curvature $1/a(z)$.

Proof. For any point with $-1 < z < 1$, choosing the argument of its first two coordinates gives a preimage under Φ . The differential computed in Lemma 8.1 is invertible between the two tangent spaces, so the inverse function theorem makes Φ a local coordinate map there. Set $E = a^2/f$ and $F = f$. The nonzero Levi-Civita coefficients of $E dv^2 + F d\theta^2$ are

$$\Gamma_{vv}^v = \frac{E'}{2E}, \quad \Gamma_{\theta\theta}^v = -\frac{F'}{2E}, \quad \Gamma_{v\theta}^\theta = \Gamma_{\theta v}^\theta = \frac{F'}{2F}.$$

Substitution in the curvature tensor gives

$$\text{Ric}_{vv} = -\frac{F''}{2F} + \frac{(F')^2}{4F^2} + \frac{E'F'}{4EF}.$$

In dimension two, contracting the curvature tensor gives $\text{Ric} = K_g g$ and scalar curvature $2K_g$. Consequently

$$K_g = -\frac{F''}{2EF} + \frac{(F')^2}{4EF^2} + \frac{E'F'}{4E^2F} = -\frac{1}{2\sqrt{EF}} \left(\frac{F'}{\sqrt{EF}} \right)'.$$

Here $\sqrt{EF} = a > 0$ and $F' = f' = -2va$, so the last expression is $1/a$. This computes the intrinsic curvature of the same metric through its actual local pullback.

The cylinder image is precisely the complement of the two poles and is dense in S^2 . Both the scalar curvature of the smooth metric and $2/a(z)$ are continuous on the whole sphere. Their equality therefore extends over the poles. The two-dimensional scalar/sectional curvature identity proves the equivalent assertions. $\square$

9 Smooth one-forms and the complete dissipation

Retain the metric constructed from a, b, c , and write $D = (a, b, c)$ for these data. Let x, y, z be the ambient coordinate functions restricted to the unit sphere. Put $\kappa = -y dx + x dy$.

Lemma 9.1. *For globally smooth $r, s : \mathbb{R} \rightarrow \mathbb{R}$, the formulas*

$$M_D(r) = a(z)r(z) dz, \quad Z_D(s) = b(z)s(z) \kappa$$

define smooth one-forms on the entire sphere. Both vanish at the poles, and their pullbacks by the actual cylinder map are

$$\Phi^* M_D(r) = a(v)r(v) dv, \quad \Phi^* Z_D(s) = f(v)s(v) d\theta. \quad (26)$$

The constant unit probes $M_D(1)$ and $Z_D(1)$ are nonzero.

Proof. Every factor is globally smooth on the sphere: the one-forms dx, dy, dz are the actual differentials of its smooth coordinate functions. At a pole $x = y = 0$, hence $\kappa = 0$; also $dz = 0$ on the tangent plane there. This proves the pole statements directly from the global definitions. The derivative formulas in Lemma 8.1 give $\Phi^* dz = dv$ and $\Phi^* \kappa = (1 - v^2)d\theta$. Using $f = (1 - v^2)b$ proves (26). At $v = 0$, evaluation on the respective cylinder coordinate directions gives $a(0) > 0$ and $f(0) > 0$ for the unit probes, proving nonvanishing. $\square$

Lemma 9.2. *For a smooth Riemannian metric g and a smooth covariant tensor T on a smooth Hausdorff manifold, its first and second total covariant derivatives are smooth and uniquely*

determined by their covariant-derivative realization laws. Both assignments are real-linear in T . For a one-form h , writing $N = \nabla h$ and $H = \nabla N$, these laws imply

$$N(X, Y) = X(h(Y)) - h(\nabla_X Y),$$

$$H(X, Y, Z) = X(N(Y, Z)) - N(\nabla_X Y, Z) - N(Y, \nabla_X Z)$$

for smooth vector fields X, Y, Z .

Proof. Use the Levi-Civita connection of the given metric. In local coordinates the covariant derivative of T is its ordinary derivative minus one Christoffel term for each covariant slot. These expressions are smooth and transform tensorially, producing the first total derivative as a global smooth tensor. Repeating the construction produces the second. The displayed formulas are its one-form and two-tensor product rules. To prove uniqueness, extend any prescribed tangent vector at a point to a global smooth vector field using a coordinate neighborhood and a smooth cutoff. The realization laws then determine every component in a tangent basis at that point. Applying the same argument first to N and then to H gives uniqueness of the pair. Real-linearity follows from the connection's linearity in its tensor argument and iteration. $\square$

On a compact smooth Riemannian surface (Σ, g) without boundary, write $K_g = \text{Scal}_g/2$ and use the trace convention

$$\Delta^\nabla h = \text{tr}_{g,12}(\nabla^2 h), \quad A_g h = \frac{1}{2}(\nabla h + (\nabla h)^\top) - \frac{1}{2} \text{tr}_g(\nabla h) g.$$

Here the first two slots of the total second derivative are traced. Thus $A_g h$ is the actual trace-free symmetric first derivative. Define the normalized complete dissipation by

$$Q_g(h) = \int_{\Sigma} (|\Delta^\nabla h|_g^2 + K_g^2 |h|_g^2 - K_g |\nabla h|_g^2 + 2K_g |A_g h|_g^2) d\mu_g. \quad (27)$$

Theorem 9.3. *For every smooth one-form, the density in (27) is smooth and integrable. Its polarization*

$$B_g(h, k) = \int_{\Sigma} (\langle \Delta^\nabla h, \Delta^\nabla k \rangle_g + K_g^2 \langle h, k \rangle_g - K_g \langle \nabla h, \nabla k \rangle_g + 2K_g \langle A_g h, A_g k \rangle_g) d\mu_g$$

is a well-defined symmetric real bilinear form, with $Q_g(h) = B_g(h, h)$. In particular, for $c \in \mathbb{R}$,

$$Q_g(ch) = c^2 Q_g(h), \quad Q_g(h \pm k) = Q_g(h) \pm 2B_g(h, k) + Q_g(k),$$

and

$$Q_g(h + k) + Q_g(h - k) = 2Q_g(h) + 2Q_g(k).$$

Proof. Lemma 9.2 supplies the actual smooth first and second derivatives. Metric contraction and permutation of covariant slots preserve smoothness. The symmetric part has the same metric trace as the original tensor; subtracting half this trace times g gives zero trace in dimension two. This proves the asserted smooth trace-free symmetric construction and its real-linearity.

The curvature and all metric inner products of smooth tensor fields are smooth. Hence each displayed density is smooth. Compactness and finiteness of Riemannian volume make these densities integrable. The derivative, rough-trace and trace-free symmetric operators are real-linear. The metric inner product is symmetric bilinear, so each of the four terms in B_g is symmetric bilinear in (h, k) . Their integrability justifies integration of the linearity identities. Evaluation at $h = k$ gives (27); expansion and scalar multiplication then give the stated quadratic identities. $\square$

Corollary 9.4. *For the produced rotational metric and every smooth one-form h ,*

$$Q_g(h) = \int_{S^2} a(z) \left(|\Delta^\nabla h|_g^2 + \frac{|h|_g^2}{a(z)^2} - \frac{|\nabla h|_g^2}{a(z)} + \frac{2|A_g h|_g^2}{a(z)} \right) d\mu_0.$$

The displayed round-measure integrand is integrable. All operators and norms in this formula belong to the same metric g .

Proof. Substitute $K_g = 1/a(z)$ from Theorem 8.3 and $d\mu_g = a(z) d\mu_0$ from Theorem 8.2. The weighted density is continuous on the compact sphere, which proves its integrability against round area. $\square$

10 The exact zonal action

Continue with a globally smooth positive balanced profile a and the same produced metric g . Write $f(v) = 2 \int_v^1 \xi a(\xi) d\xi$ and $H(v) = f(v)/a(v)$. Since a is positive on the compact interval, H is smooth on a neighborhood of $[-1, 1]$. For a globally smooth real probe r , set

$$L_a r(v) = H(v)r''(v) + (H'(v) - 2v)r'(v) - r(v), \quad (28)$$

$$J_a[r] = \int_{-1}^1 \{H(L_a r)^2 + (H - 2v^2)r^2 + 2vHrr'\} dv. \quad (29)$$

These are ordinary integrals of continuous functions on the closed interval.

Lemma 10.1. *Let $U(v) = \int_0^v a(t)r(t) dt$. The global meridional form is $M(r) = d(U \circ z)$, and its canonical derivatives satisfy*

$$\nabla M(r) = a(z)r'(z) dz \otimes dz - zr(z)g, \quad (30)$$

$$\Delta^\nabla M(r) = (L_a r)(z) dz. \quad (31)$$

In particular, everywhere on the sphere,

$$\begin{aligned} |dz|_g^2 &= f(z)/a(z)^2, & |M(r)|_g^2 &= f(z)r(z)^2, \\ \text{tr}_g \nabla M(r) &= H(z)r'(z) - 2zr(z), \\ |\nabla M(r)|_g^2 &= (H(z)r'(z) - zr(z))^2 + (zr(z))^2, \\ |A_g M(r)|_g^2 &= \frac{1}{2}(H(z)r'(z))^2. \end{aligned}$$

No equatorial symmetry of a or r is assumed.

Proof. The fundamental theorem of calculus and the chain rule give $d(U \circ z) = a(z)r(z)dz$, including at the poles. On the genuine open cylinder of Lemma 8.1, the metric is $E dv^2 + f d\theta^2$, where $E = a^2/f$ and $f' = -2va$. The Levi-Civita formula therefore gives

$$\Gamma_{vv}^v = \frac{a'}{a} + \frac{va}{f}, \quad \Gamma_{\theta\theta}^v = \frac{vf}{a}, \quad \Gamma_{v\theta}^v = 0.$$

The covariant derivative of the actual covector $ar dv$ has components $ar' - va^2r/f$, $-vfr$, and zero mixed components. This is exactly (30). Both sides are smooth global tensors; the cylinder has dense image, so the identity holds at the poles as well.

The round height gradient has squared length $1 - z^2$. The metric tensor and the pole-factor identity show that the g -dual of dz is $(b/a^2) \text{grad}_{g_0} z$. Thus $|dz|_g^2 = f/a^2$ globally. Contracting (30) gives the displayed trace and norm formulas. Equivalently, away from the poles its two eigenvalues are $Hr' - vr$ and $-vr$; subtracting half the trace leaves eigenvalues $Hr'/2$ and $-Hr'/2$. Continuity again includes both poles.

For completeness, the exact-form commutation formula with the convention $\Delta^\nabla = \text{tr}_g \nabla^2$ is

$$\Delta^\nabla dF = d(\Delta_g F) + \text{Ric}_g(\nabla F, \cdot).$$

It follows by commuting the first two covariant derivatives of dF and contracting their curvature term. All derivatives here are those of the same Levi-Civita connection. On a surface $\text{Ric}_g = K_g g$, and Theorem 8.3 gives $K_g = 1/a(z)$. Since $\Delta_g(U \circ z) = (Hr' - 2vr)(z)$, differentiating this scalar identity gives

$$\Delta^\nabla M(r) = (Hr'' + (H' - 2v)r' - 2r + r)(z) dz.$$

This is (31). □

Theorem 10.2. *For every globally smooth real probe r , the complete action of the original global meridional section is*

$$Q_g(M(r)) = 2\pi J_a[r].$$

For every constant positive profile $a \equiv c$, the fixed unit probe satisfies $J_a[1] = 4/3$ and $Q_g(M(1)) = 8\pi/3$.

Proof. Insert Lemma 10.1 into the four terms of (27) and multiply the density by a . The rough term becomes $H(L_a r)^2$, and the curvature-square term becomes Hr^2 . The two first-derivative terms together are

$$-((Hr' - vr)^2 + (vr)^2) + (Hr')^2 = 2vHrr' - 2v^2r^2.$$

Their sum is precisely the integrand in (29). The volume and height identities in Theorem 8.2 then give the factor 2π . This calculation retains all four terms and requires no endpoint integration by parts. The continuous interval density and the smooth sphere density are genuinely integrable.

If $a \equiv c > 0$, then $f = c(1 - v^2)$ and $H = 1 - v^2$. For $r \equiv 1$, $L_a r = -1$ and the density is $2 - 4v^2$. Its integral is $4/3$, giving the stated geometric normalization for the same metric and section at every positive scale c . □

Lemma 10.3. *Let g be a smooth metric on a compact smooth surface and ϕ a smooth self-diffeomorphism. Pull back a smooth covariant tensor by its actual derivative in every slot. Then*

$$\nabla^{\phi^*g}(\phi^*T) = \phi^*(\nabla^g T), \quad (\nabla^{\phi^*g})^2(\phi^*T) = \phi^*((\nabla^g)^2 T).$$

The rough Laplacian, tensor norms and trace-free symmetric projection transform under the same pullback. In particular,

$$Q_{\phi^*g}(\phi^*h) = Q_g(h), \quad B_{\phi^*g}(\phi^*h, \phi^*k) = B_g(h, k).$$

These statements require no orientation-preserving hypothesis.

Proof. The pulled-back connection is metric compatible and torsion free for ϕ^*g , so uniqueness identifies it with that metric's Levi-Civita connection. Evaluating a covariant derivative on smooth vector fields, the scalar chain rule and the connection identity transform its derivative term and every slot correction. This proves the first identity; apply it again for the second. Contraction by the inverse pulled-back metric commutes with pullback, proving the rough-trace, norm and trace-free symmetric identities. Curvature and its scalar contraction transform through this same connection. Thus each of the four density terms pulls back by composition with ϕ . Riemannian volume transforms with the absolute Jacobian, so change of variables proves the action identity for both orientations. Polarization proves the pairing identity. The tensor pullback itself satisfies the identity and contravariant composition laws by the ordinary derivative chain rule, hence also both inverse-cancellation laws. $\square$

Lemma 10.4. *For the produced rotational metric and globally smooth probes r, s ,*

$$B_g(M(r), Z(s)) = 0, \quad Q_g(M(r) + Z(s)) = Q_g(M(r)) + Q_g(Z(s)).$$

Proof. The actual sphere reflection $\rho(x, y, z) = (x, -y, z)$ preserves the round metric, height and its differential. It therefore preserves the produced metric tensor as well. Its covector pullback fixes dz and sends $\kappa = -y dx + x dy$ to $-\kappa$. Consequently $\rho^*M(r) = M(r)$ and $\rho^*Z(s) = -Z(s)$, with the derivative of ρ included in both pullbacks. Lemma 10.3 and bilinearity give $B_g(M(r), Z(s)) = -B_g(M(r), Z(s))$, hence the pairing vanishes. The quadratic expansion then gives the sum identity. Reflection fixes height rather than reversing it, so no equatorial symmetry is needed. $\square$

Lemma 10.5. *For the same produced metric, define the smooth tensor*

$$\Omega_x(V, W) = a(z(x)) \det(x, d\iota_x V, d\iota_x W),$$

where $\iota : S^2 \hookrightarrow \mathbb{R}^3$ is the actual inclusion. It is alternating, has squared tensor norm 2, and is parallel for the metric's Levi-Civita connection.

Proof. The determinant is a finite sum of products of ambient coordinate functions and their differentials, so the tensor is smooth globally. For tangent vectors V, W , the square of the round determinant form is $g_0(V, V)g_0(W, W) - g_0(V, W)^2$: expand the scalar triple product using $|x| = 1$ and $x \perp d\iota V, d\iota W$. The metric determinant identity from Theorem 8.2 multiplies this Gram determinant by $a(z)^2$. Therefore $\Omega(e_1, e_2)^2 = 1$ in every g -orthonormal basis. Alternation then gives $|\Omega|_g^2 = 2$.

Covariant differentiation preserves alternation. On a two-dimensional tangent space, every alternating two-tensor is a multiple of the nonzero tensor Ω . Metric compatibility and differentiation of its constant squared norm give $\langle \nabla_X \Omega, \Omega \rangle_g = 0$ for every tangent vector X . That multiple must vanish, proving $\nabla \Omega = 0$. $\square$

Theorem 10.6. *For the produced metric and its area form from Lemma 10.5, define the actual smooth Hodge rotation by*

$$(\mathcal{R}h)_x(Y) = \Omega_x(h_x^\sharp, Y).$$

It acts on the final covector slot of both canonical covariant derivatives and commutes with the rough Laplacian. It preserves the complete density and action of every smooth one-form. With the outward determinant convention,

$$\mathcal{R}M(r) = -Z(r), \quad Q_g(Z(r)) = 2\pi J_a[r].$$

Consequently, for every pair of globally smooth real probes r, s ,

$$Q_g(M(r) + Z(s)) = 2\pi(J_a[r] + J_a[s]). \quad (32)$$

Proof. The metric sharp map and contraction of smooth tensors produce a smooth global one-form. Metric compatibility and $\nabla\Omega = 0$ give

$$(\nabla\mathcal{R}h)(X, Y) = \Omega((\nabla_X h)^\sharp, Y).$$

To differentiate once more, extend Y to a smooth vector field and differentiate the first-slot contraction of ∇h . The derivative of that vector field occurs on both sides; subtracting the connection correction in the second derivative cancels it. The correction in the final covector slot is already part of the first-derivative formula. Thus

$$(\nabla^2\mathcal{R}h)(X, Y, Z) = \Omega((\nabla^2 h)(X, Y, \cdot)^\sharp, Z).$$

Taking the actual metric trace in X, Y proves $\Delta^\nabla\mathcal{R}h = \mathcal{R}\Delta^\nabla h$.

In an orthonormal frame put $\varepsilon = \Omega(e_1, e_2)$, so $\varepsilon^2 = 1$. The covector components (u, v) rotate to $(-\varepsilon v, \varepsilon u)$. This preserves their norm and the norm of every row of the first derivative. If a two-tensor has matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$, its trace-free symmetric part has squared norm $((A - D)^2 + (B + C)^2)/2$. Rotating its final slot replaces its rows by $(-\varepsilon B, \varepsilon A)$ and $(-\varepsilon D, \varepsilon C)$, which preserves this expression. The rough, zeroth-order, first-order and Ahlfors norms in the complete density therefore all agree before and after rotation. Integrating proves $Q_g(\mathcal{R}h) = Q_g(h)$ without a curvature-sign assumption.

For the actual sphere, the included round height gradient is $e_3 - zx$. Direct determinant expansion gives $\Omega_0(\text{grad}_{g_0} z, Y) = -\kappa(Y)$. The metric-dual formula used in Lemma 10.1 gives $M(r)^\sharp = (b/a)r \text{grad}_{g_0} z$. Since $\Omega = a\Omega_0$, it follows globally that $\mathcal{R}M(r) = -br\kappa = -Z(r)$. The quadratic action is unchanged by a minus sign, so the azimuthal action equals the meridional action. Combine Theorem 10.2 with Lemma 10.4 to obtain (32). $\square$

11 The smooth Haar projection and complete action split

Lemma 11.1. *For a unit complex number ζ , the ambient map*

$$(x, y, z) \longmapsto ((\text{Re } \zeta)x - (\text{Im } \zeta)y, (\text{Im } \zeta)x + (\text{Re } \zeta)y, z)$$

restricts to a smooth diffeomorphism of the actual unit sphere and preserves every produced metric g . Write R_θ for $\zeta = e^{i\theta}$. Then $R_0 = \text{id}$, $R_{\theta+\phi} = R_\theta \circ R_\phi$, $R_{-\theta} = R_\theta^{-1}$, and $R_{\theta+2\pi} = R_\theta$. The map $(\theta, x) \mapsto R_\theta x$ is jointly smooth, including at the poles, and its derivative in a sphere tangent direction is the restriction of the same ambient orthogonal map.

Proof. The identity $(\text{Re } \zeta)^2 + (\text{Im } \zeta)^2 = 1$ proves orthogonality; complex multiplication gives the group law. Restriction to the sphere preserves the literal map and its derivative. It fixes height and its differential and preserves the round metric, so it preserves the produced tensor $b(z)g_0 + (c(z)/b(z))dz \otimes dz$. The real-angle formula consists of sines, cosines and ambient linear coordinates, hence is jointly smooth before restriction. The sphere restriction remains smooth everywhere and has no pole singularity. $\square$

Lemma 11.2. *Let g be a fixed smooth Riemannian metric on a finite-dimensional smooth manifold without boundary. A jointly smooth real-parameter family T_t of smooth covariant tensors has a smooth fiberwise integral over every finite oriented interval $[a, b]$. Evaluation and the actual covariant derivative commute with that integral:*

$$\left(\int_a^b T_t dt \right)_x (V_1, \dots, V_s) = \int_a^b (T_t)_x (V_1, \dots, V_s) dt, \quad \nabla \int_a^b T_t dt = \int_a^b \nabla T_t dt.$$

Pairing with a fixed tensor, metric traces and the trace-free symmetric projection also commute with this integral.

Proof. Each fiber is finite dimensional, so its continuous parameter curve has a genuine Bochner integral on a compact interval. Fix a bundle trivialization near a base point. Its fiber map is continuous linear and therefore commutes with integration. In these fixed coordinates, smooth parameter integration proves smoothness of the integral section. For each derivative order, compactness of the parameter interval and joint regularity provide the local bounds needed to differentiate under the integral; no single bound for all orders is required.

Evaluate the covariant derivative on smooth vector fields. Its scalar derivative term passes through the integral by the parameter derivative formula. Each connection correction is evaluation of T_t on fixed tangent vectors at the point under consideration, so evaluation and the finite sum of corrections pass through the same integral. This proves the claimed identity for the genuine covariant derivative. Finally, metric pairing and each stated contraction are continuous linear maps on the fixed tensor fiber, which proves their integral identities. $\square$

For a covariant tensor T , the pullback R_θ^*T includes dR_θ in every covariant slot. This family is jointly smooth: the tangent map of the joint action on a vector with zero angle component is exactly the fixed-angle derivative, and smooth tensor evaluation preserves joint smoothness. We may consequently define P on all smooth covariant tensors by normalized fiberwise integration. For a one-form its literal evaluation is

$$(Ph)_x(V) = \frac{1}{2\pi} \int_0^{2\pi} h_{R_\theta x}((dR_\theta)_x V) d\theta. \quad (33)$$

Theorem 11.3. *The operator P produces smooth global tensors, is real-linear and idempotent, and satisfies $R_\phi^*PT = PT$ for every circle rotation. Moreover $PT = T$ if and only if T is invariant under that actual circle action. In particular $PM(r) = M(r)$ for every globally smooth probe; the meridional unit form is a nonzero element of the image for every produced metric.*

Proof. Smoothness and evaluation follow from Lemma 11.2. The tensor integrals are genuinely integrable, so linearity follows from ordinary integral linearity. The group law changes the integrand of $P(R_\phi^*T)$ or R_ϕ^*PT by an angular translation. Every evaluated orbit is 2π -periodic; its integral over one period is unchanged by that translation. Thus both expressions equal PT . An invariant tensor has a constant orbit, whose normalized integral equals the original tensor. Apply this fact to the already invariant PT to obtain $P^2T = PT$ and the stated characterization of its range. Every unit complex number is $e^{i\phi}$ for some real ϕ , so these identities apply to the whole circle group. The actual rotations preserve z and dz , hence fix $a(z)r(z)dz$. Its unit-probe member is nonzero by Lemma 9.1. $\square$

Theorem 11.4. *For the same produced metric, the actual first and second covariant derivatives commute with P in every tensor rank. For smooth one-forms,*

$$\Delta^\nabla Ph = P\Delta^\nabla h, \quad A_g(Ph) = P(A_g h).$$

All these statements are global and include the poles.

Proof. Every R_θ preserves g . Lemma 10.3 therefore gives $\nabla R_\theta^* T = R_\theta^* \nabla T$ for the same connection. Pass this identity through the actual integral using Lemma 11.2. Apply the result again to ∇T for the second derivative. Metric trace commutes with this integral and with isometric pullback, giving the rough-Laplacian identity. The same argument for the trace-free symmetric projection, after the first derivative, proves the Ahlfors identity. $\square$

Theorem 11.5. *For every produced metric and smooth one-forms h, k , the actual Haar projection is self-adjoint for the complete four-term polarization:*

$$B_g(Ph, k) = B_g(h, Pk), \quad B_g(Ph, k - Pk) = 0.$$

The original complete action has the exact decomposition

$$Q_g(h) = Q_g(Ph) + Q_g(h - Ph). \quad (34)$$

No curvature pinching assumption is required for these identities.

Proof. Theorem 11.4 and the linear contraction identities apply to each term of the original polarized density: the rough pairing, the K^2 -weighted form pairing, the negative K -weighted derivative pairing, and the positive $2K$ -weighted Ahlfors pairing. The resulting rotated density is jointly continuous on the angle interval times the compact sphere. Its product integral is finite, so Fubini gives exactly

$$B_g(Ph, k) = \frac{1}{2\pi} \int_0^{2\pi} B_g(R_\theta^* h, k) d\theta. \quad (35)$$

Isometry invariance of the full pairing gives $B_g(R_\theta^* h, k) = B_g(h, R_{-\theta}^* k)$. Inversion of the angle preserves integration over a full period; symmetry of B_g and (35) then give $B_g(Ph, k) = B_g(h, Pk)$. Since $P(k - Pk) = 0$ by idempotence, self-adjointness implies $B_g(Ph, k - Pk) = 0$. Expand the original quadratic action on $h = Ph + (h - Ph)$ and use this zero mixed term to obtain (34). $\square$

12 Global classification of invariant one-forms

The smoothness of the height probes at the poles is essential. The cylinder coefficients alone do not provide it. We first record the smooth factorization that removes this difficulty without assuming analyticity.

Lemma 12.1. *Every smooth even function $e : \mathbb{R} \rightarrow \mathbb{R}$ has the form $e(t) = E(t^2)$ for some globally smooth E . Every smooth odd function has the form $tF(t^2)$ for some globally smooth F . Consequently, if the smooth pair $(P, Q) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is rotation-equivariant, there are globally smooth $A, B : \mathbb{R} \rightarrow \mathbb{R}$ such that*

$$\begin{aligned} P(u, v) &= uA(u^2 + v^2) - vB(u^2 + v^2), \\ Q(u, v) &= vA(u^2 + v^2) + uB(u^2 + v^2). \end{aligned} \quad (36)$$

Proof. For a smooth odd function p , its Hadamard quotient $q(t) = \int_0^1 p'(st) ds$ is smooth and satisfies $p(t) = tq(t)$. Cancellation for $t \neq 0$, and continuity at zero, show that q is even. For smooth even e , apply this observation to the odd function e' and write $e'(t) = 2tTe(t)$ with Te smooth and even. On $s > 0$,

$$\frac{d}{ds}e(\sqrt{s}) = Te(\sqrt{s}).$$

Both sides have continuous limits at zero. The one-sided fundamental theorem of calculus gives the same derivative identity at zero. Iteration with the smooth even functions $T^j e$ proves that $s \mapsto e(\sqrt{s})$ is smooth on the closed half-line.

For completeness, this closed-half-line function has a global smooth extension. If its right-hand jets at zero are a_j , choose a smooth compactly supported cutoff η equal to one near zero and take

$$G(s) = \sum_{j=0}^{\infty} \frac{a_j}{j!} s^j \eta(\lambda_j s).$$

Choose $\lambda_j \geq 1$ successively large enough that every derivative of the j th summand of order $k < j$ has supremum norm at most 2^{-j} . This is possible since those norms scale as λ_j^{k-j} . The series converges in every fixed derivative order and has jets a_j . Joining G on the negative half-line to the given function on the nonnegative half-line gives a smooth function: at each order the two one-sided derivatives have the same limit at zero. Applying this construction to $e(\sqrt{s})$ proves the even factorization, and applying it to the even quotient q proves the odd factorization.

Restrict the equivariant pair to $(t, 0)$. Rotation by π makes both restrictions odd, hence they are $tA(t^2)$ and $tB(t^2)$. For $\rho = \sqrt{u^2 + v^2} > 0$, rotate $(\rho, 0)$ to (u, v) with cosine u/ρ and sine v/ρ . Equivariance gives (36). At the origin, both components vanish by the same oddness, so the formula holds there as well. $\square$

Lemma 12.2. For $\epsilon \in \{1, -1\}$, put $d = 4 + u^2 + v^2$ and define the actual sphere map

$$\Psi_\epsilon(u, v) = \left(\frac{4u}{d}, \frac{4v}{d}, \frac{\epsilon(4 - u^2 - v^2)}{d} \right).$$

It is smooth, intertwines plane rotation with the sphere rotation, and parametrizes the complement of the pole of height $-\epsilon$. Its inverse on that complement is

$$(x, y, z) \mapsto \left(\frac{2x}{1 + \epsilon z}, \frac{2y}{1 + \epsilon z} \right).$$

Its actual tangent map is bijective everywhere, and

$$\Psi_\epsilon^* dz = -\frac{16\epsilon}{d^2}(u du + v dv), \quad \Psi_\epsilon^* \kappa = \frac{16}{d^2}(u dv - v du). \quad (37)$$

For every smooth rotation-invariant one-form h on the sphere, there are globally smooth $A_\epsilon, B_\epsilon : \mathbb{R} \rightarrow \mathbb{R}$ for which

$$\Psi_\epsilon^* h = A_\epsilon(u^2 + v^2)(u du + v dv) + B_\epsilon(u^2 + v^2)(u dv - v du). \quad (38)$$

Proof. The displayed vector has norm one. Its positive denominator makes it a smooth sphere-valued map, and direct substitution verifies equivariance and both inverse laws. Also $1 + \epsilon z = 8/d$ on its image. For a plane tangent (U, V) , its included derivative is

$$\left(\frac{4U}{d} - \frac{8u(uU + vV)}{d^2}, \frac{4V}{d} - \frac{8v(uU + vV)}{d^2}, -\frac{16\epsilon(uU + vV)}{d^2} \right).$$

If this vector is zero, the last component first gives $uU + vV = 0$, and the first two give $U = V = 0$. Thus the actual tangent map is injective, and hence bijective between two-dimensional tangent spaces. Evaluating dz and $\kappa = -y dx + x dy$ on it proves (37), including at $(0, 0)$.

The two coefficients of $\Psi_\epsilon^* h$ are smooth because h and the actual tangent map are smooth. The chain rule and rotation invariance of h imply that their coefficient pair is rotation-equivariant: evaluation on a rotated tangent vector at the rotated point equals the original evaluation. Lemma 12.1 then gives (38) on the entire plane. $\square$

Theorem 12.3. *For the produced metric g_D , every smooth one-form invariant under its specified circle rotations can be written*

$$h = M_D(r) + Z_D(s)$$

with globally smooth real functions $r, s : \mathbb{R} \rightarrow \mathbb{R}$. The equality holds on the entire sphere, including both poles; no equatorial symmetry is assumed.

Proof. First construct global smooth coefficients P, Q with $h = P(z) dz + Q(z) \kappa$, independently of the metric. On the chart in Lemma 12.2, write $t = u^2 + v^2$. The exact identities

$$t = \frac{4(1 - \epsilon z)}{1 + \epsilon z}, \quad \frac{d^2}{16} = \frac{4}{(1 + \epsilon z)^2}$$

convert (38) into local height coefficients

$$P_\epsilon(z) = -\frac{4\epsilon}{(1 + \epsilon z)^2} A_\epsilon \left(\frac{4(1 - \epsilon z)}{1 + \epsilon z} \right), \quad Q_\epsilon(z) = \frac{4}{(1 + \epsilon z)^2} B_\epsilon \left(\frac{4(1 - \epsilon z)}{1 + \epsilon z} \right).$$

These evaluations determine the original one-form because the actual tangent map is surjective.

Here is an explicit way to extend and glue these coefficients. Choose a smooth transition S with $0 \leq S \leq 1$, $S = 0$ on $(-\infty, 0]$, $S > 0$ on $(0, \infty)$ and $S = 1$ on $[1, \infty)$. The denominator $\delta(w) = 1 + wS(4w + 3)$ is strictly positive for every real w and equals $1 + w$ for $w \geq -1/2$: for $w \leq -3/4$ it is one, while for $-3/4 < w < 0$ it is at least $1 + w > 1/4$. Replace $1 + \epsilon z$ in the preceding formulas by $\delta(\epsilon z)$. This produces globally smooth coefficients still representing h on $\epsilon z \geq -1/2$, including the corresponding pole. Blend the north and south coefficients with $S(z + 1/2)$ and $1 - S(z + 1/2)$. On the overlap both pairs represent the same original covector; outside the overlap the appropriate weight is one. Thus the global smooth P, Q satisfy $h = P(z) dz + Q(z) \kappa$ everywhere.

It remains to divide by the metric coefficients without assuming their positivity outside $[-1, 1]$. For any globally smooth c nonzero on that interval, set

$$r_c(v) = \frac{P(v)c(v)}{c(v)^2 + S(v^2 - 1)}.$$

Its denominator is positive everywhere: on the physical interval $c^2 > 0$, and outside it $S(v^2 - 1) > 0$. Hence r_c is globally smooth and $c(v)r_c(v) = P(v)$ on $[-1, 1]$. Apply this construction to $(c, P) = (a, P)$ and (b, Q) . The resulting r, s give precisely the already defined sections $M_D(r)$ and $Z_D(s)$. $\square$

Corollary 12.4. *For every smooth one-form h there are globally smooth $r, s : \mathbb{R} \rightarrow \mathbb{R}$ with*

$$Ph = M_D(r) + Z_D(s), \quad Q_{g_D}(h) = 2\pi(J_a[r] + J_a[s]) + Q_{g_D}(h - Ph).$$

Proof. The actual smooth average Ph is invariant by Theorem 11.3. Apply Theorem 12.3 to that same section, then use Theorem 10.6 and the complete split in Theorem 11.5. No sign assertion about the remainder is needed for this identity. $\square$

13 Scalarization of the complete action

For a covariant two-tensor T , write

$$A_g T = \frac{T + T^\top}{2} - \frac{\text{tr}_g T}{2} g.$$

Thus the already defined one-form operator is $A_g h = A_g(\nabla h)$. On an oriented Riemannian surface let Ω be its smooth unit area tensor, normalized by $|\Omega|_g^2 = 2$. The actual scalar contractions associated with h are

$$\tau = \text{tr}_g(\nabla h), \quad \omega = \langle \Omega, \nabla h \rangle_g.$$

In a positively oriented orthonormal frame, $\omega = \nabla_0 h_1 - \nabla_1 h_0$. Both functions are smooth, being contractions of the canonical first derivative.

Lemma 13.1. *For every smooth covariant two-tensor T on an oriented Riemannian surface, put $t = \text{tr}_g T$ and $w = \langle \Omega, T \rangle_g$. Then*

$$T = A_g T + \frac{t}{2} g + \frac{w}{2} \Omega, \quad |T|_g^2 = |A_g T|_g^2 + \frac{t^2 + w^2}{2}, \quad \langle A_g T, T \rangle_g = |A_g T|_g^2. \quad (39)$$

The genuine covariant derivatives satisfy

$$|\nabla T|_g^2 = |\nabla(A_g T)|_g^2 + \frac{|dt|_g^2 + |dw|_g^2}{2}. \quad (40)$$

Proof. At a point choose a positively oriented orthonormal basis and write $T = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$. Then $t = p + s$, $w = q - r$ and

$$A_g T = \frac{1}{2} \begin{pmatrix} p - s & q + r \\ q + r & s - p \end{pmatrix}.$$

The symmetric trace-free, scalar metric and alternating summands are mutually orthogonal, with $|g|^2 = |\Omega|^2 = 2$. This proves (39) by direct matrix calculation.

The area tensor is parallel: its derivative remains alternating, hence is pointwise proportional to Ω ; differentiation of its constant squared norm makes that proportionality coefficient zero, as in the argument of Lemma 10.5. Metric compatibility and parallelness therefore give

$$dt(X) = \text{tr}_g(\nabla_X T), \quad dw(X) = \langle \Omega, \nabla_X T \rangle_g,$$

$$\nabla_X(A_g T)(Y, Z) = \frac{(\nabla_X T)(Y, Z) + (\nabla_X T)(Z, Y)}{2} - \frac{dt(X)}{2} g(Y, Z).$$

These are the actual tensor derivative rules, including the connection correction in each covariant slot. Applying the same orthogonal matrix calculation to $\nabla_X T$ and summing over an orthonormal basis for X proves (40). $\square$

Lemma 13.2. *On a closed Riemannian manifold of positive dimension, let T and V be smooth covariant tensors of ranks s and $s + 1$. Define $\text{div}_g V$ by tracing the first two slots of ∇V . Then*

$$\int \langle \nabla T, V \rangle_g d\mu_g = - \int \langle T, \text{div}_g V \rangle_g d\mu_g.$$

Consequently

$$\int \langle \nabla T, \nabla U \rangle_g d\mu_g = - \int \langle T, \Delta^\nabla U \rangle_g d\mu_g,$$

and the actual rough Laplacian is formally self-adjoint.

Proof. Let X be the metric dual of the smooth covector $Y \mapsto \langle T, V(Y, \cdot) \rangle_g$. The metric product rule gives

$$\operatorname{div}_g X = \langle \nabla T, V \rangle_g + \langle T, \operatorname{div}_g V \rangle_g.$$

Every term is smooth and integrable on the compact manifold. Integrate and use the divergence theorem on the manifold without boundary. Taking $V = \nabla U$ gives the rough-Laplacian formula. Symmetry of the metric pairing then gives formal self-adjointness. $\square$

Write S for transposition of the first two slots of a covariant three-tensor: $(SB)(X, Y, Z) = B(Y, X, Z)$. This slot permutation is distinct from the circle-average projector P .

Lemma 13.3. *On a Riemannian surface, every smooth covariant two-tensor T satisfies*

$$\operatorname{div}_g(S\nabla T) = \nabla(\operatorname{div}_g T) + 2K_g A_g T. \quad (41)$$

If the surface is closed, then

$$\int \langle S\nabla T, \nabla T \rangle_g d\mu_g = \int |\operatorname{div}_g T|_g^2 d\mu_g - 2 \int K_g |A_g T|_g^2 d\mu_g. \quad (42)$$

Proof. The curvature convention is $R(X, Y)Z = K_g(\langle Y, Z \rangle_g X - \langle X, Z \rangle_g Y)$. In an orthonormal frame normal at the point of calculation, the difference of the two differential operators in (41) is

$$\sum_i ([\nabla_i, \nabla_j]T)_{ik} = - \sum_i T(R(e_i, e_j)e_i, e_k) - \sum_i T(e_i, R(e_i, e_j)e_k).$$

The surface curvature formula reduces this to

$$K_g(T_{jk} + T_{kj} - (\operatorname{tr}_g T)g_{jk}) = 2K_g(A_g T)_{jk}.$$

Tensoriality makes the result independent of the chosen frame. No symmetry or trace-free assumption on T was used.

Apply Lemma 13.2 first with $V = S\nabla T$, then with the lower-rank tensor $\operatorname{div}_g T$ and the tensor T . Substitute (41). The first term becomes $\int |\operatorname{div}_g T|^2$, while $\langle T, A_g T \rangle_g = |A_g T|_g^2$ by (39). This gives (42). Smoothness and compactness justify splitting the integrals. $\square$

Lemma 13.4. *For a smooth one-form on a Riemannian surface, its actual second covariant derivative $B = \nabla^2 h$ obeys*

$$|B|_g^2 - \langle SB, B \rangle_g = K_g^2 |h|_g^2.$$

Proof. For any three-tensor in an orthonormal two-frame, expansion gives

$$\sum_{i,j,k} B_{ijk}^2 - \sum_{i,j,k} B_{jik} B_{ijk} = (B_{010} - B_{100})^2 + (B_{011} - B_{101})^2.$$

For the genuine second derivative, the covector Ricci identity is

$$(\nabla_X \nabla h)(Y, Z) - (\nabla_Y \nabla h)(X, Z) = -h(R(X, Y)Z).$$

The two differences above are respectively $K_g h(e_1)$ and $-K_g h(e_0)$, so their squared sum is exactly $K_g^2 |h|_g^2$. $\square$

Theorem 13.5. *On a closed oriented Riemannian surface, every smooth one-form satisfies the following identity for the original complete action:*

$$Q_g(h) = \int |\nabla(A_g h)|_g^2 d\mu_g + 3 \int K_g |A_g h|_g^2 d\mu_g + \frac{1}{2} \left(\int (|d\tau|_g^2 - K_g \tau^2) d\mu_g + \int (|d\omega|_g^2 - K_g \omega^2) d\mu_g \right), \quad (43)$$

where $\tau = \text{tr}_g(\nabla h)$ and $\omega = \langle \Omega, \nabla h \rangle_g$ are the actual contractions. All terms are integrable. No curvature-sign or pinching hypothesis is required.

Proof. Set $N = \nabla h$ and $B = \nabla N$. The divergence of N is the actual rough Laplacian $\Delta^\nabla h$. Combining Lemmas 13.3 and 13.4 gives

$$\int (|\Delta^\nabla h|_g^2 + K_g^2 |h|_g^2) d\mu_g = \int |B|_g^2 d\mu_g + 2 \int K_g |A_g h|_g^2 d\mu_g. \quad (44)$$

Substitution into (27) yields

$$Q_g(h) = \int |B|_g^2 d\mu_g - \int K_g |N|_g^2 d\mu_g + 4 \int K_g |A_g h|_g^2 d\mu_g.$$

The two pointwise identities from Lemma 13.1 are

$$|N|_g^2 = |A_g h|_g^2 + \frac{\tau^2 + \omega^2}{2}, \quad |B|_g^2 = |\nabla(A_g h)|_g^2 + \frac{|d\tau|_g^2 + |d\omega|_g^2}{2}.$$

Inserting them proves (43). The original fields, their canonical derivatives, the curvature and the scalar contractions are smooth. Thus every displayed integrand is continuous and integrable on the compact surface, justifying all additions and subtractions. $\square$

Corollary 13.6. *For the produced metric g_D , use the actual area tensor Ω_D from Lemma 10.5, and define $\tau_D = \text{tr}_{g_D}(\nabla h)$ and $\omega_D = \langle \Omega_D, \nabla h \rangle_{g_D}$. Every smooth one-form obeys*

$$Q_{g_D}(h) = \int |\nabla(A_{g_D} h)|_{g_D}^2 d\mu_{g_D} + 3 \int \frac{|A_{g_D} h|_{g_D}^2}{a(z)} d\mu_{g_D} + \frac{1}{2} \left(\int \left(|d\tau_D|_{g_D}^2 - \frac{\tau_D^2}{a(z)} \right) d\mu_{g_D} + \int \left(|d\omega_D|_{g_D}^2 - \frac{\omega_D^2}{a(z)} \right) d\mu_{g_D} \right).$$

Proof. The constructed Ω_D is smooth, alternating and of squared norm two. Theorem 8.3 gives the same metric's curvature $K_{g_D} = 1/a(z)$ everywhere, including both poles. Apply Theorem 13.5 to that metric, that area tensor, and the original form h . $\square$

14 Scalar Haar compatibility

We use the same rotations R_θ , metric g_D and area tensor Ω_D as above. For a scalar function u on the sphere, put

$$P_0 u(x) = \frac{1}{2\pi} \int_0^{2\pi} u(R_\theta x) d\theta. \quad (45)$$

This scalar operator is distinguished from the covector pullback average P .

Lemma 14.1. *Every rotation of the actual circle action preserves the constructed area tensor: $R_\theta^* \Omega_D = \Omega_D$ on the whole sphere.*

Proof. Write L_θ for the ambient orthogonal rotation fixing the third coordinate. Its matrix has determinant one: its horizontal block has determinant $\cos^2 \theta + \sin^2 \theta = 1$ and its vertical entry is one. The derivative of R_θ is the restriction of L_θ to tangent vectors. The defining ambient formula therefore gives

$$\begin{aligned} (R_\theta^* \Omega_D)_x(V, W) &= a(z(R_\theta x)) \det(L_\theta x, L_\theta V, L_\theta W) \\ &= a(z(x)) \det(x, V, W) = (\Omega_D)_x(V, W). \end{aligned}$$

Here tangent vectors are identified with their actual ambient inclusion. The computation also holds at either pole, and equally for every unit complex number representing a circle rotation. $\square$

Lemma 14.2. *The operator P_0 maps smooth scalars to smooth scalars, is real linear on continuous scalars, fixes constants, and satisfies*

$$P_0 u(R_\phi x) = P_0 u(x), \quad P_0(u \circ R_\phi) = P_0 u, \quad P_0(P_0 u) = P_0 u.$$

At either pole p , $P_0 u(p) = u(p)$. Consequently a scalar with zero angular average vanishes at both poles.

Proof. Joint smoothness of $(\theta, x) \mapsto R_\theta x$ makes $(\theta, x) \mapsto u(R_\theta x)$ smooth when u is smooth. In any relatively compact coordinate neighborhood, every derivative with respect to x is bounded on the product with the compact angle interval. Differentiation under the integral thus proves smoothness of every order. For continuous inputs the angle integrands are genuinely integrable, giving linearity. The normalization makes the average of a constant equal to that constant.

The group law changes either rotation of the input or rotation of the base point into a shift of the angle variable. Periodicity makes the integral over one full period unchanged. The resulting averaged function is invariant, so averaging it again integrates a constant angle orbit and proves idempotence. Every rotation fixes either pole, whence its scalar orbit there is the constant $u(p)$. This proves the last assertion without division by a polar weight. $\square$

Theorem 14.3. *For the original smooth one-form h , let $\tau(h) = \text{tr}_{g_D}(\nabla h)$ and $\omega(h) = \langle \Omega_D, \nabla h \rangle_{g_D}$. Then*

$$\tau(Ph) = P_0 \tau(h), \quad \omega(Ph) = P_0 \omega(h).$$

In particular, for $h_\perp = h - Ph$,

$$P_0 \tau(h_\perp) = 0, \quad P_0 \omega(h_\perp) = 0. \tag{46}$$

Both actual scalars $\tau(h_\perp)$ and $\omega(h_\perp)$ vanish at both poles. No curvature pinching or invariance hypothesis on h is imposed.

Proof. First recall why the scalar contractions have the required naturality. If F is a diffeomorphism, an orthonormal frame for F^*g is taken by dF to an orthonormal frame for g . Computing contractions in these two frames proves

$$\text{tr}_{F^*g}(F^*T) = (\text{tr}_g T) \circ F, \quad \langle F^*S, F^*T \rangle_{F^*g} = \langle S, T \rangle_g \circ F.$$

The already established canonical derivative naturality identifies $\nabla^{F^*g}(F^*h)$ with $F^*(\nabla^g h)$, including the derivative of F in both slots. These statements hold with no orientation restriction

when the area tensor is also pulled back. For the present rotations, both g_D and Ω_D are preserved, the latter by Lemma 14.1. It follows that

$$\tau(R_\theta^* h) = \tau(h) \circ R_\theta, \quad \omega(R_\theta^* h) = \omega(h) \circ R_\theta.$$

The tensor average commutes with the actual covariant derivative. At each fixed base point, trace and contraction against the fixed area tensor are continuous linear maps on a finite-dimensional tensor fiber. They commute with its integral. All angle integrands are continuous on a compact interval, so this interchange is justified by genuine integrability. Taking those two contractions of $\nabla(Ph) = P(\nabla h)$ proves the displayed compatibility identities with exactly the normalization in (45).

The actual complementary form satisfies $P(h - Ph) = 0$. Apply compatibility to that form; the covariant derivative of the zero section is zero, so both scalar averages vanish. Finally apply Lemma 14.2 at the poles. $\square$

The pole values in this section are the first endpoint facts needed for the scalar energy estimate. We now establish the additional bounds that justify division by the vanishing polar weight.

15 The positive scalar Haar estimate

Lemma 15.1. *The actual map*

$$\Phi(s, \theta) = (\sin s \cos \theta, \sin s \sin \theta, \cos s)$$

is smooth on $\mathbb{R}^2$ with values in the unit sphere, including at $s = 0, \pi$. Its derivatives $V_s = \partial_s \Phi$ and $V_\theta = \partial_\theta \Phi$ satisfy

$$g_D(V_s, V_s) = \frac{a(\cos s)^2}{b(\cos s)}, \quad g_D(V_\theta, V_\theta) = b(\cos s) \sin^2 s, \quad g_D(V_s, V_\theta) = 0.$$

They form a tangent basis when $\sin s \neq 0$. For every smooth scalar u , writing $y(s, \theta) = u(\Phi(s, \theta))$, its actual differential therefore obeys

$$|du|_{g_D}^2(\Phi(s, \theta)) = \frac{b(\cos s)}{a(\cos s)^2} y_s^2 + \frac{y_\theta^2}{b(\cos s) \sin^2 s} \quad (\sin s \neq 0). \quad (47)$$

For every continuous scalar F on the sphere,

$$\int F d\mu_{g_D} = \int_0^\pi \int_0^{2\pi} a(\cos s) \sin s F(\Phi(s, \theta)) d\theta ds. \quad (48)$$

Proof. The displayed ambient map has norm one and is smooth, so its restriction to the sphere is a genuine smooth map. Direct differentiation gives

$$V_s = (\cos s \cos \theta, \cos s \sin \theta, -\sin s), \quad V_\theta = (-\sin s \sin \theta, \sin s \cos \theta, 0).$$

Their round Gram matrix is $\text{diag}(1, \sin^2 s)$, while $dz(V_s) = -\sin s$ and $dz(V_\theta) = 0$. Substitute these into the original tensor $b(z)g_{\text{round}} + (c(z)/b(z))dz \otimes dz$ and use $a^2 - b^2 = (1 - z^2)c$. This proves the three metric evaluations even at the poles. Away from the poles, the round Gram

matrix proves independence, and dimension two gives spanning. Inverting the resulting metric Gram matrix and applying the chain rule to $du(V_s)$ and $du(V_\theta)$ proves (47).

For completeness, the volume formula follows without treating the polar map as a global one-to-one chart. Rotations preserve g_D and its volume. Fubini, justified by continuity on the compact angle interval times the compact sphere, therefore gives $\int P_0 F d\mu_{g_D} = \int F d\mu_{g_D}$. Every point x lies on the orbit of $\Phi(\arccos z(x), 0)$: choose the argument of its horizontal complex coordinate. The same assertion holds at the poles because their horizontal coordinate is zero. Thus the invariant average is the continuous height function $H(v) = P_0 F(\Phi(\arccos v, 0))$ evaluated at $v = z(x)$. The established height disintegration gives $2\pi \int_{-1}^1 a(v) H(v) dv$. Substituting $v = \cos s$ and using the literal normalization of P_0 proves (48). All integrands used in this argument are continuous on their compact domains. $\square$

Lemma 15.2. *If $f \in C^1(\mathbb{R})$, $f(2\pi) = f(0)$ and $\int_0^{2\pi} f = 0$, then*

$$\int_0^{2\pi} f^2 d\theta \leq \int_0^{2\pi} (f')^2 d\theta.$$

Proof. With normalized Fourier coefficients $\hat{f}(n) = (2\pi)^{-1} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta$, integration by parts and the matching endpoint values give $\hat{f}'(n) = in\hat{f}(n)$. The zero mean removes $n = 0$. Parseval's identity for L^2 functions on the circle [1, Corollary 2, p. 5] gives

$$\frac{1}{2\pi} \int f^2 = \sum_{n \neq 0} |\hat{f}(n)|^2 \leq \sum_{n \in \mathbb{Z}} |\hat{f}'(n)|^2 = \frac{1}{2\pi} \int (f')^2.$$

Both functions are in L^2 because they are continuous on the compact interval. Only the endpoint values of f need agree; no endpoint condition on f' is required. The Fourier identities concern the interval representatives of the corresponding L^2 functions on the circle. $\square$

Lemma 15.3. *Let $y \in C^2(\mathbb{R}^2)$ satisfy $y(0, \theta) = y(\pi, \theta) = 0$ for every θ . On each compact angular interval there is $C > 0$ such that, for $0 \leq s \leq \pi$,*

$$|y| \leq C \sin s, \quad |y_s| \leq C, \quad |y_\theta| \leq C \sin s.$$

If $w : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and positive on $[0, \pi]$ and $W(s) = \sin s w(s)$, the following functions are integrable on every such polar rectangle:

$$W y_s^2, \quad \frac{y^2}{W}, \quad \frac{y_\theta^2}{W}, \quad \frac{(W y_s - \cos s y)^2}{W}, \quad -\sin s y^2 + 2 \cos s y y_s, \quad \frac{\sin^2 s y^2}{W}, \quad \sin s y^2, \quad \cos s y^2.$$

The quotients are considered on the open radial interval; their assigned endpoint values do not affect the integrals. The boundary primitive $\cos s y(s, \theta)^2$ tends to zero at both ends.

Proof. The first radial derivative and the mixed radial-angular derivative are bounded on the compact rectangle. The angular derivative vanishes at both radial endpoints because the two boundary slices are identically zero. Apply the mean value estimate from the nearer endpoint to y and to y_θ . On the northern half, $s \leq (\pi/2) \sin s$; on the southern half, $\pi - s \leq (\pi/2) \sin s$. Enlarging one positive constant proves all three uniform bounds.

Let $m = \min_{[0, \pi]} w > 0$. For either $v = y$ or $v = y_\theta$, the bound $|v| \leq C \sin s$ gives

$$0 \leq \frac{v^2}{\sin s w(s)} \leq \frac{C^2}{m} \quad (0 < s < \pi).$$

These measurable quotients are therefore integrable on the finite rectangle. The radial, potential and boundary terms are continuous and integrable. Multiplication of y^2/W by the bounded factors $\sin^2 s$ and $\cos^2 s$ preserves integrability. Finally,

$$\frac{(Wy_s - \cos s y)^2}{W} = Wy_s^2 + \cos^2 s \frac{y^2}{W} - 2 \cos s y y_s$$

proves integrability of the completed square. Continuity and the zero boundary values give the stated limits; the uniform bound also controls the boundary primitive by $C^2 \sin^2 s$. $\square$

Lemma 15.4. *Let $y \in C^1(\mathbb{R})$ vanish at 0 and π . Suppose $W > 0$ on $(0, \pi)$ and both Wy'^2 and y^2/W are integrable there. Then the square term below is integrable and*

$$\int_0^\pi \left(Wy'^2 + \frac{y^2}{W} - \sin s y^2 \right) ds = \int_0^\pi \frac{(Wy' - \cos s y)^2}{W} ds + \int_0^\pi \frac{\sin^2 s y^2}{W} ds. \quad (49)$$

No derivative or endpoint positivity of W is required.

Proof. Expansion of the square and $\sin^2 s + \cos^2 s = 1$ give the pointwise identity

$$Wy'^2 + \frac{y^2}{W} - \sin s y^2 = \frac{(Wy' - \cos s y)^2}{W} + \frac{\sin^2 s y^2}{W} + (\cos s y^2)'.$$

The expanded square is integrable by the two finite-energy hypotheses and continuity of y, y' . The last derivative is $-\sin s y^2 + 2 \cos s y y'$, a continuous integrable function. Its integral is zero by the fundamental theorem of calculus and the two zero boundary values. This proves the identity and its positive lower bound after dropping the square. $\square$

Theorem 15.5. *Under the hypotheses of Lemma 15.3, suppose also that $y(s, 2\pi) = y(s, 0)$ and $\int_0^{2\pi} y(s, \theta) d\theta = 0$ for every s . Then*

$$\int_0^\pi \int_0^{2\pi} \frac{\sin^2 s y^2}{W} d\theta ds \leq \int_0^\pi \int_0^{2\pi} \left(Wy_s^2 + \frac{y_\theta^2}{W} - \sin s y^2 \right) d\theta ds.$$

Proof. All terms are product integrable by Lemma 15.3. Consequently almost every radial slice has the finite energies needed for Lemma 15.4. Integrate its lower bound in θ and apply Fubini. For every interior radial point, Lemma 15.2 and $W(s) > 0$ give $\int y^2/W \leq \int y_\theta^2/W$ in the angular variable. Integrating this comparison in s and adding the common radial and potential terms proves the claim. All slice and integral exchanges use the previously proved product integrability. $\square$

Theorem 15.6. *For the original produced metric g_D , every smooth scalar u with $P_0 u = 0$ satisfies the positive estimate*

$$\int (|du|_{g_D}^2 - K_{g_D} u^2) d\mu_{g_D} \geq \int \frac{u^2}{b(z)} d\mu_{g_D}. \quad (50)$$

All polar endpoint and integrability conditions follow from these hypotheses. No curvature pinching is assumed.

Proof. Set $y = u \circ \Phi$ and $w(s) = b(\cos s)/a(\cos s)$. The actual global polar map makes y smooth on $\mathbb{R}^2$. The condition $P_0 u = 0$ gives both zero pole values by Lemma 14.2; hence y has zero radial boundary slices. Rotation compatibility of Φ gives the angular period and zero angular integrals directly from the literal definition of P_0 . Positivity and continuity of a, b on the physical height interval make w continuous and positive, with a positive minimum on $[0, \pi]$. Thus every hypothesis, bound and integrability conclusion of Theorem 15.5 applies.

By Lemma 15.1, the Dirichlet density times the volume Jacobian is $W y_s^2 + y_\theta^2 / W$, where $W = \sin s b(\cos s)/a(\cos s)$. The actual curvature identity $K_{g_D} = 1/a(z)$ changes its potential density into $\sin s y^2$. The right side of (50) has polar density $a(\cos s) \sin s y^2 / b(\cos s) = \sin^2 s y^2 / W$. The exact volume formula and Theorem 15.5 therefore give (50). The intrinsic densities are continuous on the compact sphere, and the polar densities are integrable by the established endpoint estimates. No expression is evaluated by division at a geometric pole. $\square$

16 Strict positivity of the nonzonal remainder

Lemma 16.1. *Let (M, g) be a compact smooth Riemannian manifold, let $w : M \rightarrow \mathbb{R}$ be continuous and strictly positive, and let T be a smooth covariant tensor. Then*

$$\int_M w |T|_g^2 d\mu_g = 0 \iff T = 0.$$

Likewise, for every continuous real scalar v , $\int_M w v^2 d\mu_g = 0$ if and only if $v = 0$.

Proof. The Riemannian density is positive in every coordinate neighborhood, so every nonempty open set has positive volume. The displayed integrands are continuous, nonnegative and integrable by compactness. A zero integral makes the integrand zero almost everywhere. Continuity and positivity of the measure on open sets then make it zero everywhere: a positive value would persist on a nonempty open neighborhood and contradict almost-everywhere vanishing. Strict positivity of w and positive definiteness of the tensor norm give $T = 0$. The scalar case uses the same argument with v^2 . Both converse implications are immediate. $\square$

Theorem 16.2. *For every produced rotational metric g_D and smooth one-form h , put $k = h - Ph$, $A = A_{g_D} k$, $\tau = \text{tr}_{g_D}(\nabla k)$ and $\omega = \langle \Omega_D, \nabla k \rangle_{g_D}$. The original complete action satisfies*

$$Q_{g_D}(k) \geq 3 \int K_{g_D} |A|_{g_D}^2 d\mu_{g_D} + \frac{1}{2} \left(\int \frac{\tau^2}{b(z)} d\mu_{g_D} + \int \frac{\omega^2}{b(z)} d\mu_{g_D} \right) \geq 0. \quad (51)$$

Moreover,

$$Q_{g_D}(h - Ph) = 0 \iff h - Ph = 0. \quad (52)$$

Thus every nonzero Haar remainder has strictly positive action, with no curvature pinching assumption.

Proof. The actual scalars τ, ω are smooth and have zero angular mean by Theorem 14.3. Apply Theorem 15.6 to each. The literal scalarization in Corollary 13.6 gives

$$Q_{g_D}(k) = \int |\nabla A|_{g_D}^2 d\mu_{g_D} + 3 \int K_{g_D} |A|_{g_D}^2 d\mu_{g_D} + \frac{1}{2} (E(\tau) + E(\omega)),$$

where $E(v) = \int (|dv|_{g_D}^2 - K_{g_D} v^2) d\mu_{g_D}$ denotes the scalar energy appearing in that identity. The two proved scalar bounds and the nonnegative first integral prove (51). Here $K_{g_D} = 1/a(z) > 0$ and $b(z) > 0$ on the entire sphere.

Suppose $Q_{g_D}(k) = 0$. All three weighted integrals in the lower bound must vanish. Apply Lemma 16.1 with weights K_{g_D} and $1/b(z)$ to obtain $A = 0$, $\tau = 0$ and $\omega = 0$ everywhere. The actual tensor decomposition from Lemma 13.1 then gives

$$\nabla k = A + \frac{\tau}{2} g_D + \frac{\omega}{2} \Omega_D = 0.$$

This is an equality of the original smooth tensor sections. Differentiating it gives $\nabla^2 k = 0$, and tracing gives $\Delta^\nabla k = 0$. Substitution in the original four-term action (27), not merely its scalar lower bound, leaves

$$0 = Q_{g_D}(k) = \int K_{g_D}^2 |k|_{g_D}^2 d\mu_{g_D}.$$

The weight $K_{g_D}^2$ is continuous and strictly positive. A final application of Lemma 16.1 gives $k = 0$. Conversely the canonical derivatives of the zero section and all four original density terms vanish, so its action is zero. Nonnegativity and the zero classification give strict positivity when $k \neq 0$. $\square$

Corollary 16.3. *For every smooth one-form on the same produced metric,*

$$Q_{g_D}(Ph) \leq Q_{g_D}(h).$$

Equality holds exactly when $h = Ph$, equivalently when h is invariant under every rotation of the actual circle action.

Proof. The complete Haar split of Theorem 11.5 gives $Q_{g_D}(h) - Q_{g_D}(Ph) = Q_{g_D}(h - Ph)$. Apply Theorem 16.2. The fixed points of the actual projector are precisely the invariant forms, by Theorem 11.3. $\square$

Corollary 16.4. *For every smooth h there are globally smooth real probes r, s such that*

$$Ph = M_D(r) + Z_D(s), \quad Q_{g_D}(h) \geq 2\pi(J_a[r] + J_a[s]).$$

For these same probes, equality holds if and only if $h = M_D(r) + Z_D(s)$.

Proof. Use the original invariant-form classification to represent the actual Ph . The exact zonal identity gives $Q_{g_D}(Ph) = 2\pi(J_a[r] + J_a[s])$ for those probes. Both the bound and its equality case follow from Corollary 16.3. The global smoothness and the section equality retain their validity at both poles; no equatorial symmetry is assumed. $\square$

17 Hemisphere substitution and conformal-Killing safety

Write $K_{g_D} = \text{Scal}_{g_D}/2$ and retain the original $H = f/a$, sections M_D, Z_D , projector P and complete action Q_{g_D} . A smooth one-form is conformal Killing, abbreviated CK, when its actual Ahlfors derivative vanishes: $A_{g_D} h = \text{sym}(\nabla h) - \frac{1}{2} \text{tr}_{g_D}(\nabla h) g_D = 0$.

Lemma 17.1. Suppose $0 < m \leq a(v) \leq M$ on $[-1, 1]$ and put $C = M/m$. The functions

$$k_+(v) = \log \frac{M}{a(v)}, \quad k_-(v) = \log \frac{M}{a(-v)}$$

are continuous on $[0, 1]$, differentiable on $(0, 1)$ and admissible at cap C . Their derivatives are $-a'(v)/a(v)$ and $a'(-v)/a(-v)$, respectively. Moreover,

$$\mathcal{I}(k_+) = \int_0^1 H(v) dv, \quad \mathcal{I}(k_-) = \int_0^1 H(-v) dv, \quad \mathcal{I}(k_+) + \mathcal{I}(k_-) = \int_{-1}^1 H(v) dv. \quad (53)$$

No equatorial symmetry is assumed.

Proof. The bounds imply $M > 0$, $C \geq 1$ and $1 \leq M/a(\pm v) \leq C$. Monotonicity of the logarithm gives both obstacle bounds. Positivity on the physical interval, smoothness of a and the chain rule give the stated regularity and derivatives; global positivity outside that interval is unnecessary. Fubini on the triangular domain in the definition of $\mathcal{I}$ gives, for every continuous k on $[0, 1]$,

$$\mathcal{I}(k) = \int_0^1 e^{k(v)} \left(\int_v^1 2se^{-k(s)} ds \right) dv.$$

The integrand is bounded on the compact square, which supplies genuine integrability before interchanging integrals. For k_+ its inner expression becomes $2 \int_v^1 sa(s) ds/a(v) = H(v)$.

For the south, let $\tilde{a}(s) = a(-s)$. Changing variables and using the original balance condition gives

$$2 \int_v^1 s\tilde{a}(s) ds = -2 \int_{-1}^{-v} ta(t) dt = 2 \int_{-v}^1 ta(t) dt = f(-v).$$

Thus its quotient is $H(-v)$. Splitting the full height integral at zero and reflecting the negative interval proves the last identity. All quotients are continuous on their compact integration intervals. $\square$

Lemma 17.2. For every $M > 0$ and real constants c, d ,

$$Q_{g_D}(M_D(c) + Z_D(d)) = 2\pi \left(2(\mathcal{I}(k_+) + \mathcal{I}(k_-)) - \frac{4}{3} \right) (c^2 + d^2), \quad (54)$$

where $k_{\pm}(v) = \log(M/a(\pm v))$. Here constant real numbers denote the corresponding globally constant smooth probes.

Proof. For a constant probe, the actual operator in (28) is $L_a c = -c$. Its action density in (29) is therefore $(2H(v) - 2v^2)c^2$. The compact-interval continuity of H supplies integrability, and $\int_{-1}^1 v^2 dv = 2/3$ yields

$$J_a[c] = \left(2 \int_{-1}^1 H(v) dv - \frac{4}{3} \right) c^2.$$

Apply the exact zonal formula of Theorem 10.6 and the hemisphere identities proved above. Those identities only need $M > 0$; the additional m, M bounds in Lemma 17.1 are used for admissibility, not this equality. $\square$

Theorem 17.3. *Define*

$$C_{\text{CK}} = \left(\frac{5 + \sqrt{13}}{3} \right)^{1/3} \exp \left(\frac{4 + 2\sqrt{13}}{9} \right) > 1. \quad (55)$$

For the actual optimal value $m(C) = \mathcal{I}(k_{*,C}) = 2/3 - \beta(C) + 1/(3\beta(C))$ from Theorem 1.1, one has

$$m(C_{\text{CK}}) = \frac{1}{3}, \quad m(C) \geq \frac{1}{3} \iff C \leq C_{\text{CK}} \quad (C \geq 1).$$

Every admissible profile continuous on $[0, 1]$ and differentiable on $(0, 1)$ at a cap $1 \leq C \leq C_{\text{CK}}$ satisfies $\mathcal{I}(k) > 1/3$.

Proof. Since $\beta > 0$, the equation $m(C) = 1/3$ is equivalent to $3\beta^2 - \beta - 1 = 0$. Its positive root is $\beta_* = (1 + \sqrt{13})/6$. The identity $\beta^2 = q^3/(q^3 + 2)$ from (5) gives $q_*^3 = (5 + \sqrt{13})/3$. Substitution in the actual cap equation (1) yields (55). Conversely this positive $q_* > 1$ satisfies the cap equation for that displayed number. Uniqueness of the canonical cap parameter therefore identifies it with q_* and proves the claimed critical value.

The canonical parameter increases strictly with C . The function $q \mapsto q^3/(q^3 + 2)$ increases strictly for $q > 0$, so the positive upper contact also increases strictly. Both terms $-\beta$ and $1/(3\beta)$ decrease strictly. Thus the actual $m(C)$ is strictly decreasing, proving the equivalence around its critical value. For $C > 1$, the contact derivative jumps in Theorem 3.2 and optimizer uniqueness exclude equality for a differentiable competitor; hence $\mathcal{I}(k) > m(C) \geq 1/3$. For $C = 1$, the obstacle box forces $k = 0$ on $[0, 1]$ and the directly computed value is $2/3 > 1/3$. This branch does not use a contact jump. $\square$

Theorem 17.4. *For g_D , every constant zonal form $M_D(c) + Z_D(d)$ is CK. A smooth invariant one-form is CK if and only if it has this representation for real constants c, d . The actual projector preserves CK, and therefore every smooth CK form h satisfies*

$$Ph = M_D(c) + Z_D(d)$$

for some real constants. The original unit meridional form $M_D(1)$ is nonzero and CK.

Proof. The CK equation is linear. The actual first derivative, metric trace and Ahlfors projection commute with derivative pullback through an isometry, by Lemma 10.3. The meridional derivative formula in Lemma 10.1 gives

$$|A_{g_D} M_D(r)|^2 = \frac{1}{2} (Hr')^2.$$

Parallel Hodge rotation preserves the actual Ahlfors norm, so the same formula holds for $Z_D(r)$. Constant probes thus give CK forms.

Apply Theorem 12.3 to an invariant smooth h , obtaining $h = M_D(r) + Z_D(s)$ with globally smooth probes. Meridian reflection is an actual isometry fixing $M_D(r)$ and negating $Z_D(s)$. Half the sum and half the difference of h and its reflected pullback are therefore CK, so each summand is CK. At every height $v \in (-1, 1)$, use a genuine cylinder point in the norm formula above. Since $H(v) > 0$, it forces $r'(v) = s'(v) = 0$. The mean-value theorem makes both probes constant on the open interval, and continuity includes its endpoints. Their values outside $[-1, 1]$ are irrelevant to the section, so this gives exactly $h = M_D(r(0)) + Z_D(s(0))$, without claiming

global constancy of the extensions. Conversely these constant sums are invariant by the actual rotation laws and CK by the first part.

The actual derivative and Ahlfors intertwining in Theorem 11.4 sends $A_{g_D}h = 0$ to $A_{g_D}(Ph) = 0$. The average is invariant, so its classification applies. The previously proved equatorial evaluation of $M_D(1)$ is nonzero; the constant-probe calculation proves its CK equation. $\square$

Lemma 17.5. *For positive real numbers κ, Λ , the actual curvature bounds $\kappa \leq K_{g_D}(x) \leq \Lambda$ for every sphere point x are equivalent to*

$$\frac{1}{\Lambda} \leq a(v) \leq \frac{1}{\kappa} \quad (-1 \leq v \leq 1).$$

Proof. Theorem 8.3 gives $K_{g_D} = 1/a(z)$ at every point, including both poles. Each height $v \in [-1, 1]$ is realized by the actual polar point $\Phi(\arccos v, 0)$. The result follows in both directions by taking reciprocals of positive quantities. No coordinate restriction removes a pole. $\square$

Theorem 17.6. *Let g_D be an original produced metric and h any smooth CK one-form. If $0 < m \leq a \leq M$ on $[-1, 1]$ and $M/m \leq C_{CK}$, then*

$$Q_{g_D}(h) \geq 0, \quad Q_{g_D}(h) = 0 \iff h = 0. \quad (56)$$

The same conclusions hold if $\kappa > 0$, $\kappa \leq K_{g_D} \leq \Lambda$ everywhere and $\Lambda/\kappa \leq C_{CK}$.

Proof. Use Lemma 17.1 at $C = M/m$. Both actual hemisphere logarithms are differentiable admissible competitors. Theorem 17.3 gives $\mathcal{I}(k_+) > 1/3$ and $\mathcal{I}(k_-) > 1/3$, including when $C = 1$. Consequently the exact coefficient $2(\mathcal{I}(k_+) + \mathcal{I}(k_-)) - 4/3$ in (54) is strictly positive. Every constant zonal sum therefore has nonnegative action; if its action is zero, then $c^2 + d^2 = 0$, hence $c = d = 0$ and the section is zero.

By Theorem 17.4, the actual Ph is such a constant sum. The full split and the original remainder theorem give

$$Q_{g_D}(h) = Q_{g_D}(Ph) + Q_{g_D}(h - Ph) \geq 0.$$

If the sum vanishes, both terms vanish. The preceding strict coefficient makes $Ph = 0$, while Theorem 16.2 makes $h - Ph = 0$. Thus $h = 0$. This argument also covers forms whose Haar average was zero from the outset. Conversely the zero section has zero original action.

For the curvature formulation, evaluating the nonempty sphere gives $\Lambda \geq \kappa > 0$. Lemma 17.5 supplies $m = 1/\Lambda$, $M = 1/\kappa$, and $M/m = \Lambda/\kappa$. Apply the profile conclusion to these actual bounds. $\square$

Corollary 17.7. *Under the profile bounds of Theorem 17.6, $Q_{g_D}(M_D(1)) > 0$, including at the critical cap.*

Proof. The unit meridional section is nonzero and CK by Theorem 17.4. Nonnegativity and the zero classification in Theorem 17.6 imply strict positivity. $\square$

18 Smooth sharpness and fixed-probe recovery

Lemma 18.1. *Let $a : \mathbb{R} \rightarrow \mathbb{R}$ be uniformly continuous and even, with $\ell \leq a \leq u$ everywhere. There exist globally smooth even functions $a_n : \mathbb{R} \rightarrow \mathbb{R}$ with the same bounds $\ell \leq a_n \leq u$ and $a_n \rightarrow a$ uniformly on $\mathbb{R}$. The endpoints ℓ, u may coincide and need not be positive.*

Proof. Choose nonnegative normalized smooth bump kernels ρ_n , supported within distance $r_n = 1/(n+1)$ of zero, with $\int \rho_n = 1$. Put $b_n = \rho_n * a$. Each convolution is smooth: locally in its argument, every derivative of the compactly supported kernel is integrated against the continuous, hence locally integrable, function a on a fixed compact set. The nonnegative mass-one kernel makes $b_n(x)$ an average of numbers in the closed convex interval $[\ell, u]$, so that interval is preserved exactly. Uniform continuity and support shrinking to zero give

$$|b_n(x) - a(x)| \leq \sup_{|y| \leq r_n} |a(x-y) - a(x)| \rightarrow 0$$

uniformly in x . Finally set $a_n(x) = (b_n(x) + b_n(-x))/2$. This preserves smoothness and the interval, gives evenness, and still converges uniformly because $a(-x) = a(x)$. $\square$

For a continuous profile positive on $[-1, 1]$, write

$$T(a) = \int_{-1}^1 \frac{f_a(v)}{a(v)} dv, \quad f_a(v) = 2 \int_v^1 sa(s) ds.$$

This is a scalar integral, not a smooth-metric construction for arbitrary continuous a .

Lemma 18.2. *Suppose a, b are continuous on $[-1, 1]$ and belong to a common positive box $0 < \ell \leq a, b \leq u$. If $|a - b| \leq \varepsilon$ there, then*

$$|T(a) - T(b)| \leq 8 \left(\frac{1}{\ell} + \frac{u}{\ell^2} \right) \varepsilon. \quad (57)$$

In particular, uniform convergence $a_n \rightarrow a$ on that interval with the same positive box and continuous a_n implies $T(a_n) \rightarrow T(a)$. No derivative convergence or balance is required.

Proof. The primitives defining f_a, f_b are continuous, and positive denominators make their quotients continuous and integrable on the closed interval. For $v \in [-1, 1]$ the elementary integral estimates give $|f_b(v)| \leq 4u$ and $|f_a(v) - f_b(v)| \leq 4\varepsilon$. Here $\varepsilon \geq 0$ follows from the assumed error bound at any point. The exact quotient identity

$$\frac{f_a}{a} - \frac{f_b}{b} = \frac{f_a - f_b}{a} + \frac{f_b(b-a)}{ab}$$

therefore bounds the pointwise absolute difference by $4\varepsilon/\ell + 4u\varepsilon/\ell^2$. Integrating on an interval of length two proves (57).

For the limit assertion, uniform convergence gives continuity of a and passes the closed box to a , so the quantitative bound applies. There is also a direct dominated-convergence proof: first pass to the limit in the defining warp integrals, dominated by u , then in their quotients, dominated by $4u/\ell$. These bounds establish actual integrability before either passage to the limit. $\square$

Lemma 18.3. For $C \geq 1$, let $p(t) = \max(0, \min(1, t))$ and define the actual extended reciprocal optimizer by

$$a_{0,C}(v) = \exp(-k_{*,C}(p(|v|))). \quad (58)$$

It is uniformly continuous and even on $\mathbb{R}$. Its bounds and physical formula are

$$\begin{aligned} \frac{1}{C} &\leq a_{0,C}(v) \leq 1 & (v \in \mathbb{R}), \\ a_{0,C}(v) &= \exp(-k_{*,C}(|v|)) & (-1 \leq v \leq 1). \end{aligned}$$

It is balanced and satisfies

$$T(a_{0,C}) = 2m(C), \quad 2\pi \left(2T(a_{0,C}) - \frac{4}{3} \right) = 8\pi \left(m(C) - \frac{1}{3} \right). \quad (59)$$

Proof. The interval projection p and the absolute-value map are Lipschitz. The continuous function $t \mapsto \exp(-k_{*,C}(t))$ on the compact interval $[0, 1]$ is uniformly continuous. Their composition has the asserted regularity; its formula is even and the optimizer's obstacle bounds give the exact reciprocal box. On the physical interval the projection is the identity. Beyond its endpoints it gives a constant extension. The function $va_{0,C}(v)$ is odd and continuous, so its integral on $[-1, 1]$ is zero.

For any globally continuous, physically positive, balanced a , the proof of Lemma 17.1 uses no derivatives or smooth metric. In particular the reflected-tail law follows from balance, all warp quotients are integrable, and

$$T(a) = \mathcal{I} \left(\log \frac{M}{a(v)} \right) + \mathcal{I} \left(\log \frac{M}{a(-v)} \right) \quad (M > 0).$$

Apply this with $a = a_{0,C}$ and $M = 1$. On $[0, 1]$, its northern logarithm is exactly $k_{*,C}$; evenness identifies the southern logarithm with it. The functional depends only on this interval, proving $T(a_{0,C}) = 2m(C)$. The second identity in (59) is algebra. No smooth metric is assigned to the nonsmooth limiting optimizer. $\square$

Theorem 18.4. For every $C \geq 1$ there are globally smooth even profiles a_n and genuine produced pole data D_n such that

$$\begin{aligned} D_n \cdot a &= a_n, & \frac{1}{C} &\leq a_n(v) \leq 1 \quad (v \in \mathbb{R}), & \int_{-1}^1 va_n(v) dv &= 0, \\ a_n &\longrightarrow a_{0,C} \text{ uniformly on } \mathbb{R}. \end{aligned}$$

For their original metrics and the fixed scalar probe $r = 1$,

$$Q_{g_{D_n}}(M_{D_n}(1)) \longrightarrow 8\pi \left(m(C) - \frac{1}{3} \right). \quad (60)$$

Proof. Apply Lemma 18.1 to $a_{0,C}$ with box $[1/C, 1]$. Each approximant is even, so the exact balance integral vanishes by oddness. Its positive lower bound permits the existing smooth pole-factor and metric construction, yielding D_n with the explicit equation $D_n \cdot a = a_n$.

For each of these actual smooth metrics, the accepted meridional reduction and constant-probe calculation give

$$Q_{g_{D_n}}(M_{D_n}(1)) = 2\pi \left(2T(a_n) - \frac{4}{3} \right).$$

Lemma 18.2 passes to the limit using the common positive box. Equation (59) identifies that limit and proves (60). Only the scalar probe is kept fixed; the actual metric and its canonical meridional section are both produced from a_n . $\square$

Theorem 18.5. *For every prescribed $C > C_{CK}$, set $C_0 = (C + C_{CK})/2$. There are a globally smooth even balanced profile a and actual produced data D with $D.a = a$, $1/C_0 \leq a \leq 1$ on $\mathbb{R}$, and*

$$1 \leq K_{g_D} \leq C_0 < C$$

on the sphere. The original section $M_D(1)$ is nonzero, is CK, and has $Q_{g_D}(M_D(1)) < 0$.

Proof. The chosen cap satisfies $1 < C_{CK} < C_0 < C$. Theorem 17.3 gives $m(C_0) < 1/3$. Apply Theorem 18.4 at C_0 . Its limit is negative, so an actual smooth member has negative action. Retain that member's profile and producer equation $D.a = a$. Lemma 17.5 converts its exact reciprocal box to the displayed curvature bounds, including both poles. The actual unit meridional section is nonzero and CK by Theorem 17.4. $\square$

Corollary 18.6. *There is a sequence as in Theorem 18.4 at $C = C_{CK}$ with $1 \leq K_{g_{D_n}} \leq C_{CK}$, each $M_{D_n}(1)$ nonzero and CK, and*

$$0 < Q_{g_{D_n}}(M_{D_n}(1)) \longrightarrow 0.$$

The profiles converge uniformly to the same $a_{0,C_{CK}}$ and retain their exact box, evenness, balance and producer equations.

Proof. The critical value is $m(C_{CK}) = 1/3$, so the action limit in (60) is zero. Each member's curvature bounds follow from its reciprocal box, and its nonzero CK property follows from the actual constant-probe theorem. Corollary 17.7 proves that every smooth action is strictly positive. No scaling of the scalar probe is used. $\square$

Theorem 18.7. *For an original produced metric with $0 < m \leq a \leq M$ on $[-1, 1]$, set $C = M/m$, $k(v) = \log(M/a(v))$ and $S = c^2 + d^2 > 0$. Define*

$$B = \frac{Q_{g_D}(M_D(c) + Z_D(d))}{4\pi S} + \frac{2}{3} - 2m(C). \quad (61)$$

Then $B = \mathcal{I}(k(v)) + \mathcal{I}(k(-v)) - 2m(C)$ exactly, and

$$\begin{aligned} \int_{-1}^1 (k(v) - k_{*,C}(|v|))^2 dv &\leq B/c(C), \\ \int_{-1}^1 (k(v) - k(-v))^2 dv &\leq 4B/c(C), \\ \int_{-1}^1 \left(k(v) - \frac{k(v) + k(-v)}{2} \right)^2 dv &\leq B/c(C). \end{aligned}$$

Here $c(C) > 0$ is the explicit stability constant from Theorem 5.2. If $S = 0$, then $c = d = 0$, the actual zonal section and its action are zero; no normalized profile conclusion is asserted.

Proof. The profile bounds imply $M > 0$, $C \geq 1$, and continuity and the exact obstacle bounds $0 \leq k \leq \log C$ on the full interval. Divide (54) by the nonzero quantity $4\pi S$ and rearrange to obtain the stated identity for B . Apply all three estimates of Corollary 6.1 to this literal full logarithm. This preserves their factor four for reflection error and requires no equatorial symmetry. For $S = 0$, nonnegativity of squares forces both coefficients to vanish. The actual section is then zero and the exact constant-zonal formula gives zero action, without any division by S . $\square$

19 The represented metric class and the sharp CK threshold

Let $\mathcal{G}_{\text{rot}}$ be the class of actual smooth metrics g on S^2 for which there are produced data D and a smooth diffeomorphism $F : S^2 \rightarrow S^2$ with $g = F^*g_D$. All tensor pullbacks below use the actual derivative of this same map. No orientation condition is imposed on F . This definition does not assume a representation theorem for arbitrary abstract effective circle actions.

Lemma 19.1. *Every produced metric belongs to $\mathcal{G}_{\text{rot}}$, and this class is closed under further smooth diffeomorphic pullback. For a fixed representation $g = F^*g_D$, put $\Omega_F = F^*\Omega_D$. Then*

$$K_g(x) = K_{g_D}(F(x)), \quad F_*\mu_g = \mu_{g_D}, \quad |\Omega_F|_g^2 = 2, \quad \nabla^g \Omega_F = 0. \quad (62)$$

The same positive curvature box holds for g and g_D . The transported curl satisfies

$$\text{curl}_{g, \Omega_F}(F^*h) = \text{curl}_{g_D, \Omega_D}(h) \circ F.$$

Proof. The identity diffeomorphism represents g_D . If $g = G^*g_D$, then $F^*g = (G \circ F)^*g_D$, proving closure with an actual representing map. The actual derivative pullback has the two inverse laws $(F^{-1})^*F^* = \text{id}$ and $F^*(F^{-1})^* = \text{id}$ for tensors and metrics. The curvature naturality and metric-volume change of variables used in Lemma 10.3 give the first two identities in (62). Surjectivity of F gives both directions of the curvature-box equivalence.

The area tensor evaluates by $\Omega_F(x)(v, w) = \Omega_D(F(x))(dF_x v, dF_x w)$. Actual tensor-norm and covariant-derivative naturality therefore transport its squared norm and parallelness. Contracting the actual first derivative against this same transported area gives the stated curl law. For an orientation-reversing map we still use Ω_F , rather than asserting that a separately fixed area orientation was preserved. Metric volume is the positive Riemannian measure, so its change of variables requires no orientation-preserving premise. $\square$

Fix $g = F^*g_D$. Write ρ for the native circle rotations and define

$$\rho_F(z) = F^{-1} \circ \rho(z) \circ F, \quad P_F A = F^* P((F^{-1})^* A). \quad (63)$$

The second expression is a smooth covariant tensor of the same rank as A .

Theorem 19.2. *The maps ρ_F form a genuine circle action by isometries of g . Its angle parametrization is jointly smooth on $\mathbb{R} \times S^2$ and has period 2π . For every smooth covariant tensor A of rank s , including $s = 0$, every $x \in S^2$ and tuple $v_1, \dots, v_s \in T_x S^2$,*

$$(P_F A)_x(v_1, \dots, v_s) = \frac{1}{2\pi} \int_0^{2\pi} A_{\rho_F(e^{i\theta})x}(d\rho_F(e^{i\theta})_x v_1, \dots, d\rho_F(e^{i\theta})_x v_s) d\theta. \quad (64)$$

The integrand is genuinely integrable. The projector is real-linear, idempotent and invariant under ρ_F , and $P_F A = A$ if and only if A is invariant under every element of ρ_F . It intertwines the original projector by $P_F(F^ A) = F^*(P A)$.*

Proof. The identity, multiplication and inverse laws follow by canceling the adjacent $F \circ F^{-1}$ in the explicit compositions. The same cancellation and $\rho(z)^*g_D = g_D$ give $\rho_F(z)^*g = g$. Joint smoothness follows by composing the jointly smooth original angle action with $(\theta, x) \mapsto (\theta, F(x))$ and F^{-1} ; its period is unchanged. The actual tangent maps consequently preserve the inner product of this same pulled-back metric.

Tensor pullback reverses composition, so for every tensor rank

$$\rho_F(z)^*A = F^*\rho(z)^*(F^{-1})^*A.$$

Evaluate the definition of P_FA at x on the prescribed slot tuple, then apply the original integral formula (33) at $F(x)$ on $(dF_x v_i)_i$. The displayed pullback identity identifies each integrand with the derivative of the actual map $\rho_F(z)$, giving (64). Joint smoothness of the tensor orbit, proved from the genuine family derivative, gives continuity at each fixed fiber and slot tuple, hence integrability on the compact angle interval.

Linearity and the two inverse pullback laws give the asserted intertwining and idempotence. Pulling the original invariant average through F gives $\rho_F(z)^*P_FA = P_FA$. Conversely invariance of A becomes invariance of $(F^{-1})^*A$ under ρ ; the original fixed-point theorem then gives $P_FA = A$. All these equations retain the same map and original tensor. $\square$

Theorem 19.3. *For a fixed representation $g = F^*g_D$, P_F commutes with the actual first and second covariant derivatives, rough Laplacian and Ahlfors projection in their respective tensor ranks. It is self-adjoint for the complete four-term pairing of g . For every original smooth one-form η ,*

$$Q_g(\eta) = Q_g(P_F\eta) + Q_g(\eta - P_F\eta), \quad Q_g(\eta - P_F\eta) \geq 0, \quad (65)$$

and $Q_g(\eta - P_F\eta) = 0$ if and only if $\eta - P_F\eta = 0$. Thus $Q_g(P_F\eta) \leq Q_g(\eta)$, with equality precisely for forms invariant under ρ_F . No pinching assumption is required. For every $g \in \mathcal{G}_{\text{rot}}$, one actual representation supplies these conclusions simultaneously for every η .

Proof. Put $h = (F^{-1})^*\eta$, so $\eta = F^*h$ by the actual inverse law. The canonical naturality from Lemma 10.3, followed by Theorem 11.4, gives, for example,

$$\nabla^g(P_FA) = F^*\nabla^{g_D}P((F^{-1})^*A) = F^*P\nabla^{g_D}((F^{-1})^*A) = P_F(\nabla^g A).$$

Iterating this equality gives the actual second derivative. The accepted rough-trace and Ahlfors naturality give the other commutations with all covariant slots retained.

Complete-action and complete-pairing naturality transport the self-adjoint identity and full Haar split from g_D . In particular

$$(F^{-1})^*(\eta - P_F\eta) = h - Ph, \quad Q_g(\eta - P_F\eta) = Q_{g_D}(h - Ph).$$

Theorem 16.2 supplies nonnegativity and zero exactly when $h - Ph = 0$. Injectivity of derivative pullback makes this equivalent to the original remainder being zero. The split then gives the comparison and its fixed-point equality, and Theorem 19.2 identifies those fixed points. Choosing a representation from the definition of the class, before choosing η , proves the final full-class assertion. $\square$

Corollary 19.4. *Fix any representation $g = F^*g_D$ and put $M_F(r) = F^*M_D(r)$, $Z_F(s) = F^*Z_D(s)$. For globally smooth real probes,*

$$Q_g(M_F(r) + Z_F(s)) = 2\pi(J_a[r] + J_a[s]).$$

For every original smooth η , there are globally smooth r, s with $P_F\eta = M_F(r) + Z_F(s)$ and

$$Q_g(\eta) = 2\pi(J_a[r] + J_a[s]) + Q_g(\eta - P_F\eta) \geq 2\pi(J_a[r] + J_a[s]).$$

For these same probes, equality holds if and only if $\eta = M_F(r) + Z_F(s)$. These statements hold for every represented metric, with its actual representation retained.

Proof. Apply complete-action naturality to the actual sum $M_D(r) + Z_D(s)$ and use Theorem 10.6. Classify $P((F^{-1})^*\eta)$ by Theorem 12.3, then pull that exact section equality through F . This gives the globally smooth probes and their stated original-section equation, including both transported poles. The full split and zero-remainder theorem prove the remaining conclusions, always using those same probes. No uniqueness of their extensions outside the physical interval is claimed. $\square$

Theorem 19.5. Let $g \in \mathcal{G}_{\text{rot}}$ and let η be any original smooth CK one-form. If $\kappa > 0$, $\kappa \leq K_g \leq \Lambda$ everywhere and $\Lambda/\kappa \leq C_{\text{CK}}$, then

$$Q_g(\eta) \geq 0, \quad Q_g(\eta) = 0 \iff \eta = 0.$$

In particular every nonzero such form has strictly positive action, including at the critical cap. For any representing D, F , the actual P_F preserves the CK equation, and $P_F\eta = M_F(c) + Z_F(d)$ for some real constants c, d .

Proof. Choose $g = F^*g_D$ and put $h = (F^{-1})^*\eta$. The actual derivative/Ahlfors pullback law makes h CK for g_D . Lemma 19.1 supplies the same curvature box on g_D , so Theorem 17.6 applies. Complete-action naturality gives $Q_g(\eta) = Q_{g_D}(h)$, and inverse pullback makes $h = 0$ equivalent to $\eta = 0$. This proves the full arbitrary-form conclusion, with no nonzero-average hypothesis.

The Ahlfors-derivative commutation in Theorem 19.3 sends its zero equation to the zero equation of $P_F\eta$. Alternatively transport the original CK Haar preservation theorem. The original constant-probe classification of Ph , pulled back through F , gives the last section equality. $\square$

Theorem 19.6. Fix $g = F^*g_D$, use its profile height $z_F = z \circ F$, and suppose $0 < m \leq a \leq M$ on $[-1, 1]$. Set $C = M/m$, $k(v) = \log(M/a(v))$ and $S = c^2 + d^2 > 0$. Then

$$B_F = \frac{Q_g(M_F(c) + Z_F(d))}{4\pi S} + \frac{2}{3} - 2m(C) = \mathcal{I}(k(v)) + \mathcal{I}(k(-v)) - 2m(C), \quad (66)$$

and all three inequalities in Theorem 18.7 hold with B_F in place of B . The reflection factor is four. For $S = 0$, both coefficients, the pulled-back section and its action vanish, without a normalized profile conclusion. Every represented metric has one representation supporting these conclusions for all profile bounds and coefficient pairs simultaneously.

Proof. The section is literally $F^*(M_D(c) + Z_D(d))$. Pullback naturality identifies its complete action with the original constant-zonal action on g_D . Substitute that equality into the exact budget identity and all three inequalities of Theorem 18.7. The profile logarithm is the same generating function in height parameter v ; on the represented sphere it is evaluated at z_F . For zero coefficient sum use the original zero-coefficient law and the linearity and zero preservation of derivative pullback. Fixing the representation before the parameters proves the final assertion. $\square$

Define $\mathcal{S}_{\text{CK}}$ to consist of the real numbers $C \geq 1$ such that $Q_g(\eta) \geq 0$ for every $g \in \mathcal{G}_{\text{rot}}$, every actual curvature box $\kappa > 0$, $\kappa \leq K_g \leq \Lambda$ with $\Lambda/\kappa \leq C$, and every original smooth CK one-form η . Define the universal CK threshold as $\sup \mathcal{S}_{\text{CK}}$.

Theorem 19.7. *The actual universal safe-cap set is*

$$\mathcal{S}_{\text{CK}} = [1, C_{\text{CK}}], \quad \sup \mathcal{S}_{\text{CK}} = \left(\frac{5 + \sqrt{13}}{3} \right)^{1/3} \exp \left(\frac{4 + 2\sqrt{13}}{9} \right). \quad (67)$$

It is nonempty and bounded above. Every smooth nonzero CK form in a represented critical safe box has positive action, while the same fixed unit probes of Corollary 18.6 lie in this class and have positive actions tending to zero. The supercritical witnesses of Theorem 18.5 also lie in this class with their exact producer and curvature equations.

Proof. If $1 \leq C \leq C_{\text{CK}}$, every curvature-box ratio at most C is at most C_{CK} . Theorem 19.5 therefore proves $C \in \mathcal{S}_{\text{CK}}$ with all the stated universal quantifiers. Conversely let $C > C_{\text{CK}}$ and choose exactly $C_0 = (C + C_{\text{CK}})/2$. The already constructed smooth witness has $1 \leq K_{g_D} \leq C_0 < C$, a nonzero CK unit meridional section, and negative original action. Its metric belongs to $\mathcal{G}_{\text{rot}}$ through the identity diffeomorphism. Applying the proposed safety property with $\kappa = 1$ and $\Lambda = C_0$ is a contradiction. Caps below one are excluded by definition, proving the set equality.

Since $C_{\text{CK}} > 1$, the interval is nonempty and bounded above by its included right endpoint. The real supremum laws and (55) give the second equality in (67). The universal strict positivity follows from Theorem 19.5. For recovery, retain exactly the profiles, produced metrics and fixed probes from the earlier critical sequence and add their identity representations. This neither constructs a second sequence nor changes their action limit. The same identity embedding retains the original supercritical witness equations. $\square$

20 Normalized area geometry and complete action realization

Lemma 20.1. *For a continuous real function u on $\mathbb{R}$ and a fixed $a \in \mathbb{R}$, put $S(x) = \int_a^x (x - t)u(t) dt$. Then*

$$S(x) = x \int_a^x u(t) dt - \int_a^x tu(t) dt, \quad S'(x) = \int_a^x u(t) dt.$$

If u is globally smooth, both its first primitive and S are globally smooth.

Proof. All integrands are continuous on every compact integration interval, so linearity gives the first identity for either orientation of the interval. Differentiate its two ordinary primitives using the fundamental theorem of calculus. The two terms $xu(x)$ cancel, leaving the stated derivative. For smooth u , the first primitive is differentiable with smooth derivative u , and S is differentiable with that smooth first primitive as derivative. Induction on the derivative order proves global smoothness. $\square$

For a curvature coefficient $K : \mathbb{R} \rightarrow \mathbb{R}$, define the actual normalized primitives

$$V_K(x) = -1 + \int_0^x K(t) dt, \quad f_K(x) = 2x - 2 \int_0^x (x - t)K(t) dt, \quad f_{1,K}(x) = 2 - 2 \int_0^x K(t) dt. \quad (68)$$

Theorem 20.2. *Suppose K is continuous on $\mathbb{R}$, strictly positive on $[0, 1]$, and satisfies the two exact normalized moments*

$$\int_0^1 K(x) dx = 2, \quad \int_0^1 xK(x) dx = 1.$$

The actual derivative identities are

$$V'_K = K, \quad f'_K = f_{1,K} = -2V_K, \quad f'_{1,K} = f''_K = -2K.$$

They hold on $\mathbb{R}$. Moreover,

$$V_K(0) = -1, \quad V_K(1) = 1, \quad f_K(0) = f_K(1) = 0, \quad f_{1,K}(0) = 2, \quad f_{1,K}(1) = -2.$$

The map V_K is a strictly increasing bijection from $[0, 1]$ onto $[-1, 1]$, and $\int_0^1 V_K = 0$. The warp f_K is strictly concave on $[0, 1]$ and strictly positive on $(0, 1)$. If K is globally smooth, all three primitives are globally smooth.

Proof. Apply Lemma 20.1 and the fundamental theorem to the literal definitions (68). These also give $f_{1,K} = -2V_K$ by algebra. The left endpoints follow directly. The zeroth moment gives the right height and slope, while

$$f_K(1) = 2 - 2 \left(\int_0^1 K(t) dt - \int_0^1 tK(t) dt \right) = 0.$$

Positive $V'_K = K$ gives strict monotonicity. Continuity, the endpoint values and the intermediate value theorem give the asserted bijection. Integrating $V_K = -f'_K/2$ between the two zero warp endpoints gives its zero integral, with integrability supplied by continuity.

The actual second derivative is strictly negative on $[0, 1]$, so f_K is strictly concave there. For $0 < x < 1$, strict concavity between its two zero endpoint values gives $f_K(x) > 0$. This step uses positive curvature; the two moment equations alone do not imply it. Smoothness follows from the last part of Lemma 20.1. $\square$

Lemma 20.3. *Let $a < b$ and let $V : \mathbb{R} \rightarrow \mathbb{R}$ be globally smooth with $V' > 0$ on $[a, b]$. There exists a globally smooth real function X such that*

$$X([V(a), V(b)]) \subset [a, b], \quad X(V(x)) = x \quad (a \leq x \leq b), \quad V(X(v)) = v \quad (V(a) \leq v \leq V(b)).$$

It has $X(V(a)) = a$, $X(V(b)) = b$, and $X'(v)V'(X(v)) = 1$ on the entire closed target interval, including both endpoints. The conclusion does not require X to invert V outside those intervals.

Proof. Strict monotonicity and the intermediate value theorem produce the actual closed-interval bijection. Its continuous inverse follows from compactness of the source and the Hausdorff property of the target. Extend this inverse continuously to the line by clamping its input to the target interval.

At a target point v , including an endpoint, put $x = V^{-1}(v)$. Since $V'(x) > 0$, the local smooth inverse theorem gives a smooth inverse branch near v . Continuity of the clamped inverse and the original inverse equation show that it agrees with this branch within the target interval near v . Thus the actual inverse is smooth on that closed interval, with its one-sided endpoint jets.

A smooth function on a nondegenerate closed real interval has a global smooth extension agreeing on the interval. One construction realizes each endpoint jet (c_j) by a cutoff series $\sum_j c_j(t - t_0)^j \chi((t - t_0)/\varepsilon_j)/j!$, where χ is smooth, compactly supported and equal to one near zero. Choose $\varepsilon_j > 0$ small enough that the C^k norm of its j th term is at most 2^{-j} for every $k < j$. Every derivative series then converges uniformly after finitely many initial terms and has exactly the prescribed jet at t_0 . Glue these left and right realizations to the given function; matching all one-sided derivatives makes the result globally smooth. Apply this extension construction to the actual interval inverse.

The extension retains both inverse equations and endpoints. Differentiate $V(X(v)) = v$ within the nondegenerate closed target interval. Uniqueness of the derivative within that interval, also at its endpoints, gives $V'(X(v))X'(v) = 1$. This identifies the ambient derivative of the global extension without imposing any exterior inverse law. $\square$

For actual primitives $f = f_K$, $f_1 = f_{1,K}$ and a smooth real probe r , define the complete normalized area action by

$$A_K(r) = \int_0^1 \left[f(fr'' + 2f_1r' - Kr)^2 - Kf_1r \left(fr' + \frac{f_1r}{2} \right) + K^2fr^2 \right] dx. \quad (69)$$

For continuous K and smooth r , every factor in this density is continuous, so the integral is genuine.

Lemma 20.4. *Let K, r be globally smooth, with $K > 0$ on $[0, 1]$ and $\int_0^1 K = 2$. Suppose D is original produced data and R is globally smooth, with the physical equations*

$$D.a(V_K(x)) = 1/K(x), \quad R(V_K(x)) = r(x) \quad (0 \leq x \leq 1).$$

Write $a = D.a$ and $H = \text{warp}(a)/a$ for the original height-profile quantities. On the entire closed area interval,

$$\begin{aligned} \text{warp}(a)(V_K(x)) &= f_K(x), & a'(V_K(x)) &= -K'(x)/K(x)^3, \\ R'(V_K(x)) &= r'(x)/K(x), & R''(V_K(x)) &= r''(x)/K(x)^2 - r'(x)K'(x)/K(x)^3, \\ H(V_K(x)) &= f_K(x)K(x), & H'(V_K(x)) &= f_{1,K}(x) + f_K(x)K'(x)/K(x). \end{aligned}$$

Consequently the actual operator from (28) satisfies

$$L_a R(V_K(x)) = \frac{f_K(x)r''(x) + 2f_{1,K}(x)r'(x) - K(x)r(x)}{K(x)}. \quad (70)$$

Proof. The actual height derivative is $V'_K = K$. The derivative of $\text{warp}(a)(V_K(x))$ is

$$-2V_K(x)a(V_K(x))K(x) = -2V_K(x) = f'_K(x).$$

Both functions vanish at $x = 0$, using $V_K(0) = -1$ and the original balance equation of D . Equality of these actual derivatives and continuity give the warp identity on the full interval.

Differentiate the coefficient and probe equations within $[0, 1]$. Since both sides have actual ambient derivatives at every point, uniqueness within the nondegenerate interval gives the displayed first derivatives, including the endpoints. Differentiate the first probe derivative equation once more to obtain the stated second derivative. Each division is justified by $K(x) > 0$.

The value of H follows from the coefficient and warp equations. Substitution in the accepted formula $H'(v) = -2v - H(v)a'(v)/a(v)$ gives its displayed derivative, since $f_{1,K} = -2V_K$. Finally substitute these actual jets into $L_a R$. The two terms involving $f_K r' K'/K^2$ cancel and the remaining terms give (70). No independent first or second jet is supplied. $\square$

Theorem 20.5. *Under the hypotheses of Lemma 20.4, the original smooth metric and section satisfy*

$$Q_{g_D}(M_D(R)) = 2\pi A_K(r).$$

The identity retains every term of the original complete action.

Proof. Let $j_a[R]$ denote the original meridional density in (29). Multiply its value at $V_K(x)$ by the actual Jacobian $V'_K(x) = K(x)$. Lemma 20.4 and $-2V_K = f_{1,K}$ give the literal identity

$$j_a[R](V_K(x))K(x) = f_K(f_K r'' + 2f_{1,K} r' - Kr)^2 - K f_{1,K} r \left(f_K r' + \frac{f_{1,K} r}{2} \right) + K^2 f_K r^2.$$

It holds at both endpoints as well. In particular no endpoint term is discarded. Positive curvature and the zeroth moment map $V_K([0, 1])$ into $[-1, 1]$, with endpoints $-1, 1$. The original density is continuous on that physical height interval. The interval change-of-variables theorem therefore applies with continuity only on the actual image; no exterior positivity of $D.a$ is needed. It identifies $\int_{-1}^1 j_a[R]$ with (69). The original meridional reduction then supplies the unchanged factor 2π . $\square$

Theorem 20.6. *Let $K, r : \mathbb{R} \rightarrow \mathbb{R}$ be globally smooth, suppose $K > 0$ on $[0, 1]$, and impose the two moments $\int_0^1 K = 2$, $\int_0^1 xK = 1$. There exist a globally smooth X , original produced data D , and a globally smooth R with*

$$V_K(0) = -1, \quad V_K(1) = 1, \quad X(-1) = 0, \quad X(1) = 1,$$

both physical interval mapping laws, both inverse laws $X(V_K(x)) = x$ on $[0, 1]$ and $V_K(X(v)) = v$ on $[-1, 1]$, and $X'(v)K(X(v)) = 1$ on the closed height interval. They satisfy $R = r \circ X$ globally and, for every $x \in [0, 1]$,

$$D.a(V_K(x)) = \frac{1}{K(x)}, \quad R(V_K(x)) = r(x), \quad \text{warp}(D.a)(V_K(x)) = f_K(x). \quad (71)$$

All coefficient/probe derivative equations in Lemma 20.4 hold, including the endpoints. For every actual sphere point p ,

$$K_{g_D}(p) = K(X(z(p))), \quad Q_{g_D}(M_D(R)) = 2\pi A_K(r). \quad (72)$$

Proof. Apply Lemma 20.3 to the actual globally smooth primitive V_K on $[0, 1]$. Its derivative is $K > 0$ and its endpoints are $-1, 1$, so the resulting X has all the stated coordinate laws. The smooth coefficient $K \circ X$ is nonzero on $[-1, 1]$. The smooth quotient extension used in Theorem 12.3 supplies a globally smooth a with $(K \circ X)a = 1$ there. Thus $a > 0$ on the physical interval and $a(V_K(x)) = 1/K(x)$. Put $R = r \circ X$.

The actual change of variables and the physical reciprocal equation yield

$$\int_{-1}^1 va(v) dv = \int_0^1 V_K(x)a(V_K(x))K(x) dx = \int_0^1 V_K(x) dx = 0.$$

Continuity establishes every integral and the last equality follows from the two moments by Theorem 20.2. The existing smooth positive balanced-profile construction therefore produces D with the exact equation $D.a = a$. Both input compatibility equations for Lemma 20.4 have

now been constructed, not assumed. That lemma and Theorem 20.5 supply the warp, actual derivative and original-action identities.

Finally $z(p) \in [-1, 1]$, so $x = X(z(p))$ belongs to $[0, 1]$ and $V_K(x) = z(p)$. The original intrinsic curvature law gives $K_{g_D}(p) = 1/D \cdot a(z(p)) = K(x)$, including both poles. The chosen maps, profile and probe remain tied in all these equations. $\square$

Corollary 20.7. *For $K = 2$, $r = 1$, one has $f_K(x) = 2x - 2x^2$, $f_{1,K}(x) = 2 - 4x$ and $A_K(1) = 4/3$. The original constant-profile metric with $D \cdot a = 1/2$ has a nonzero unit meridional section and*

$$Q_{g_D}(M_D(1)) = 2\pi A_2(1) = 8\pi/3.$$

Proof. Evaluate the literal primitives at the constant curvature two. The complete area density becomes $32x - 32x^2 - 4$, whose integral from zero to one is $4/3$. For the actual constant produced data D with profile $1/2$, the physical equations are $D \cdot a(V_2(x)) = 1/2$ and $1 = 1$. Theorem 20.5 gives the exact geometric identity, and the accepted equatorial evaluation makes its unit meridional form nonzero. $\square$

20.1 The critical curvature in normalized area coordinates

Lemma 20.8. *Let $a : \mathbb{R} \rightarrow \mathbb{R}$ be continuous on $\mathbb{R}$ and strictly positive on $[-1, 1]$. Set*

$$L_a = \int_{-1}^1 a(v) dv, \quad X_a(v) = \frac{1}{L_a} \int_{-1}^v a(s) ds.$$

Then $L_a > 0$ and X_a maps $[-1, 1]$ continuously and strictly increasingly onto $[0, 1]$, with $X_a(-1) = 0$, $X_a(1) = 1$ and $X'_a = a/L_a$. Let V_a be the physical inverse, extended constantly to -1 and 1 outside $[0, 1]$, and put $K_a(x) = L_a/a(V_a(x))$. The functions V_a, K_a are continuous on $\mathbb{R}$, $K_a > 0$, and

$$\begin{aligned} V_a(X_a(v)) &= v & (-1 \leq v \leq 1), \\ X_a(V_a(x)) &= x & (0 \leq x \leq 1), \\ K_a(X_a(v)) &= \frac{L_a}{a(v)} & (-1 \leq v \leq 1). \end{aligned}$$

If $\int_{-1}^1 va(v) dv = 0$, then

$$\begin{aligned} \int_0^1 K_a &= 2, & \int_0^1 xK_a(x) dx &= 1, \\ f_{K_a}(X_a(v)) &= \frac{\text{warp}(a)(v)}{L_a}, & f'_{K_a}(X_a(v)) &= -2v \quad (-1 \leq v \leq 1). \end{aligned} \tag{73}$$

Under the same balance hypothesis, the complete area action satisfies

$$A_{K_a}(1) = 2 \int_{-1}^1 \frac{\text{warp}(a)(v)}{a(v)} dv - \frac{4}{3}. \tag{74}$$

Proof. Positivity gives $L_a > 0$ and $X'_a > 0$ on the physical interval. Compactness and strict monotonicity give the homeomorphism onto its endpoint interval; clamping the inverse gives the stated continuous extension. In particular its range remains in $[-1, 1]$, so its curvature is continuous and positive on the whole line. All inverse identities above concern the indicated physical intervals.

The chain rule and the curvature tying equation give $(V_{K_a} \circ X_a)' = K_a(X_a)a/L_a = 1$. Both sides agree at $v = -1$, hence $V_{K_a}(X_a(v)) = v$. Thus $f'_{K_a}(X_a(v)) = -2v$ and the zeroth moment is two. The derivative of $f_{K_a} \circ X_a$ is $-2va/L_a$, which equals the derivative of $\text{warp}(a)/L_a$. Balance makes both functions zero at -1 , proving the warp equation. At $v = 1$ the warp vanishes; the corrected triangular primitive, together with the zeroth moment, then gives the first moment.

For the unit probe the full density reduces to $2K_a^2 f_{K_a} - K_a(f'_{K_a})^2/2$. After multiplication by the actual Jacobian $X'_a = a/L_a$, it becomes $2\text{warp}(a)/a - 2v^2$ on $[-1, 1]$. Continuity gives integrability, and ordinary interval substitution gives (74). This calculation requires no derivative of a or K_a and assigns no smooth metric to these continuous data. $\square$

Theorem 20.9. *Take $a_* = a_{0,C_{CK}}$ from (58) and use the preceding construction to define L_*, X_*, V_*, K_* . Then K_* has both normalized moments, is continuous and positive, and obeys*

$$L_* \leq K_*(x) \leq L_* C_{CK}, \quad L_* < 2 < L_* C_{CK}.$$

On $[0, 1]$ it is symmetric about $1/2$, nonincreasing on $[0, 1/2]$ and nondecreasing on $[1/2, 1]$. There are $0 < l < u < 1$ such that $K_ = L_*$ on $[l, u]$, and $f_{K_*} > 0$ throughout $(0, 1)$. The actual complete unit-probe area action is zero:*

$$A_{K_*}(1) = 0. \tag{75}$$

Proof. The actual reciprocal optimizer is even, continuous, lies in $[1/C_{CK}, 1]$, equals one on its positive central contact interval and equals $1/C_{CK}$ at the outer contacts. Its logarithmic profile is nondecreasing on the northern half, so the reciprocal is nonincreasing there. The strict inequalities for L_* follow by integrating the two box bounds: each is strict on a nonempty interval because $C_{CK} > 1$ and both contact regions are nonempty. Evenness gives balance and $X_*(-v) = 1 - X_*(v)$, whence $X_*(0) = 1/2$ and $V_*(1-x) = -V_*(x)$ on $[0, 1]$. These identities prove the symmetry of K_* , and the increasing inverse transports the two hemisphere monotonicity laws. A closed subinterval of the central contact region maps to an interior constant-curvature plateau. Positive curvature and the two moments give strict interior warping positivity by Theorem 20.2. Finally the critical variational value is $m(C_{CK}) = 1/3$, and the reciprocal warp-ratio integral is twice that value. Substitution in (74) gives $2(2/3) - 4/3 = 0$. $\square$

Lemma 20.10. *Let $K : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, and fix $0 < l < u < 1$ with $K = \kappa$ on (l, u) . Write $f = f_K$, and let χ be globally smooth with support compactly contained in (l, u) . The complete action has the exact variation*

$$A_K(1 - \varepsilon\chi) = A_K(1) - 2\varepsilon\kappa^2 \int_0^1 f\chi + \varepsilon^2 A_K(\chi). \tag{76}$$

Proof. Expansion of the full six-entry density gives the coefficient of $-\varepsilon$

$$-2Kf^2\chi'' - 5Kff'\chi' + (4K^2f - K(f')^2)\chi.$$

On the support of χ , use $f'' = -2\kappa$ to write this as

$$2\kappa^2 f\chi + (-2\kappa f^2\chi' - \kappa f f'\chi)'.$$

Off that support, χ and both derivatives vanish, so the equality remains valid. The flux vanishes at both endpoints. All terms are continuous, and integrating the identity proves (76), with its quadratic term equal to the actual action of the same bump. $\square$

Theorem 20.11. *The actual critical curvature K_* admits a globally smooth probe r which equals one near both endpoints, is nonconstant on $[0, 1]$, has $r(0) = 1$, $r'(0) = 0$, and satisfies $A_{K_*}(r) < 0$.*

Proof. Choose a nonnegative, nonzero smooth bump χ supported strictly inside the positive plateau. Since $f_{K_*} > 0$ there, its weighted pairing is positive. Set

$$A = L_*^2 \int_0^1 f_{K_*} \chi > 0, \quad B = A_{K_*}(\chi), \quad \varepsilon = \frac{A}{|B| + 1} > 0, \quad r = 1 - \varepsilon \chi.$$

Then $\varepsilon|B| < A$, and therefore $-2\varepsilon A + \varepsilon^2 B < 0$. Equations (75) and (76) give the negative action. The same probe has the actual derivatives $r' = -\varepsilon \chi'$ and $r'' = -\varepsilon \chi''$. Its support condition gives constancy near both endpoints and the stated initial data; positivity of the pairing ensures physical nonconstancy. Only continuity of the curvature has been used; smooth geometric recovery is a subsequent step. $\square$

21 Smooth recovery and strict threshold separation

Lemma 21.1. *For $n \geq 1$ and a real function q , set*

$$B_n q(y) = \sum_{j=0}^n \binom{n}{j} y^j (1-y)^{n-j} q(j/n).$$

This is a polynomial on $\mathbb{R}$. On $[0, 1]$ it preserves every closed box containing the sampled values, and it is nondecreasing if q is nondecreasing there. If q is bounded on $[0, 1]$, then $B_n q(y) \rightarrow q(y)$ at each point where the restriction of q to $[0, 1]$ is continuous, including either endpoint with relative continuity.

For a real function K , define

$$\rho(x) = (2x - 1)^2, \quad q_K(y) = K\left(\frac{1 + \sqrt{y}}{2}\right), \quad S_n K(x) = B_n q_K(\rho(x)).$$

The function $S_n K$ is globally smooth and satisfies $S_n K(1 - x) = S_n K(x)$ for every real x . If K is nondecreasing on $[1/2, 1]$, then $S_n K$ is nondecreasing there and nonincreasing on $[0, 1/2]$. A box for K on $[0, 1]$ is preserved by $S_n K$ on $[0, 1]$. If also $K(1 - x) = K(x)$ for $x \in [0, 1]$, then $q_K(\rho(x)) = K(x)$ on that interval.

Proof. The Bernstein basis is nonnegative and sums to one on $[0, 1]$, proving box preservation. The exact adjacent-difference derivative formula is

$$(B_{n+1} q)'(y) = (n+1) \sum_{j=0}^n \binom{n}{j} y^j (1-y)^{n-j} \left(q\left(\frac{j+1}{n+1}\right) - q\left(\frac{j}{n+1}\right) \right).$$

Thus monotone sampled values give a nonnegative derivative. For convergence at a relative continuity point y , subtract $q(y)$ inside the convex sum. Split the nodes into those within distance δ of y and their complement. The first part is bounded by the local oscillation. If $|q(z) - q(y)| \leq D$ on the interval, the complementary part is at most

$$\frac{D}{\delta^2} \sum_{j=0}^n (j/n - y)^2 \binom{n}{j} y^j (1-y)^{n-j} = \frac{D y(1-y)}{n \delta^2} \rightarrow 0.$$

This proves the asserted pointwise convergence without assuming global continuity of q .

The radial argument lies in $[1/2, 1]$ when $y \in [0, 1]$. Both it and ρ on $[1/2, 1]$ are nondecreasing, giving the right-hand order law. Since $\rho(1-x) = \rho(x)$, reflection gives the other order law. The polynomial composition gives global smoothness. Finally $\sqrt{(2x-1)^2} = |2x-1|$; the radial argument is x on the right half and $1-x$ on the left, proving the reconstruction identity. $\square$

Theorem 21.2. *Let $\ell < m < u$. Suppose $K : \mathbb{R} \rightarrow \mathbb{R}$ has a strongly measurable restriction to $[0, 1]$ and satisfies*

$$\ell \leq K(x) \leq u, \quad K(1-x) = K(x) \quad (0 \leq x \leq 1), \quad \int_0^1 K = m,$$

and is nondecreasing on $[1/2, 1]$. There are functions $G_n : \mathbb{R} \rightarrow \mathbb{R}$, each smooth on $\mathbb{R}$, such that for every n their physical restrictions are symmetric, nonincreasing on $[0, 1/2]$, nondecreasing on $[1/2, 1]$, and obey

$$\ell \leq G_n(x) \leq u \quad (0 \leq x \leq 1), \quad \int_0^1 G_n = m, \quad \int_0^1 x G_n(x) dx = \frac{m}{2}. \quad (77)$$

Moreover $\int_0^1 |G_n - K| \rightarrow 0$, and there is a strictly increasing subsequence index ν with $G_{\nu(n)}(x) \rightarrow K(x)$ for almost every $x \in [0, 1]$.

Proof. We first construct an approximant with any prescribed L^1 error $\varepsilon > 0$. Choose

$$0 < \eta < \min \left(1, \frac{\varepsilon}{u - \ell} \right), \quad K_\eta = (1 - \eta)K + \eta m, \quad \delta = \eta \min(m - \ell, u - m) > 0.$$

This preserves symmetry, hemisphere order and mean m , and gives the strict box $\ell + \delta \leq K_\eta \leq u - \delta$ on $[0, 1]$. Also $\int_0^1 |K_\eta - K| \leq \eta(u - \ell) < \varepsilon$.

Apply Lemma 21.1 to K_η . The radial profile q_{K_η} is monotone on $[0, 1]$, so its relative discontinuity set is countable. The inverse image of this set under ρ is countable: ρ is injective separately on each closed hemisphere. Hence $S_n K_\eta \rightarrow K_\eta$ almost everywhere on $[0, 1]$. The unchanged strict box bounds the absolute difference by its width; dominated convergence gives L^1 convergence and convergence of the means.

Put

$$c_n = m - \int_0^1 S_n K_\eta, \quad H_n = S_n K_\eta + c_n.$$

Then $c_n \rightarrow 0$ and $\int_0^1 H_n = m$ exactly. For large n , $|c_n| < \delta$, so H_n stays in the original box $[\ell, u]$ on the physical interval. Adding the constant changes neither smoothness nor reflection nor either hemisphere order law. Furthermore

$$\int_0^1 |H_n - K| \leq \int_0^1 |S_n K_\eta - K_\eta| + |c_n| + \eta(u - \ell).$$

Choose the degree after fixing η and δ , making the right side less than ε . Reflection gives the other exact moment: substitution $x \mapsto 1-x$ implies $2 \int_0^1 x H_n(x) dx = \int_0^1 H_n = m$. All integral operations are justified by measurability and the fixed finite physical box.

Choose such approximants for errors $1/(n+1)$. Their errors tend to zero, proving the sequence assertion. Finally choose strictly increasing indices whose L^1 errors have finite sum. Tonelli's theorem makes the sum of the pointwise absolute errors finite almost everywhere, and therefore those errors tend to zero. Every selected member retains (77). $\square$

Lemma 21.3. *Let $K_n, K : \mathbb{R} \rightarrow \mathbb{R}$ be continuous on $\mathbb{R}$, and fix one globally smooth $r : \mathbb{R} \rightarrow \mathbb{R}$. Suppose that $|K_n(x)| \leq B$ for every n and $x \in [0, 1]$, that $\int_0^1 |K_n - K| \rightarrow 0$, and that $K_n \rightarrow K$ almost everywhere on $[0, 1]$. Then*

$$\begin{aligned} |f_{K_n}(x) - f_K(x)| &\leq 2 \int_0^1 |K_n - K|, \\ |f_{1,K_n}(x) - f_{1,K}(x)| &\leq 2 \int_0^1 |K_n - K| \quad (0 \leq x \leq 1). \end{aligned} \tag{78}$$

Thus both primitives converge uniformly on $[0, 1]$, and the complete actions of this same probe satisfy $A_{K_n}(r) \rightarrow A_K(r)$.

Proof. Subtract the actual primitive formulas (68). On $0 \leq s \leq x \leq 1$, the kernel $|x - s|$ is at most one. The integral triangle inequality and monotonicity of the integral of $|K_n - K|$ give (78); the right sides do not depend on x .

The same estimates against zero give $|f_{K_n}|, |f_{1,K_n}| \leq 2 + 2B$ on the physical interval. The three fixed functions r, r', r'' are bounded there by compactness. Choose a common nonnegative bound M for all six entries $f_{K_n}, f_{1,K_n}, K_n, r, r', r''$. For the full density

$$D(f, f_1, K, r, r_1, r_2) = f(fr_2 + 2f_1r_1 - Kr)^2 - Kf_1r(fr_1 + f_1r/2) + K^2fr^2,$$

the triangle and product inequalities give

$$|D| \leq M(M^2 + 2M^2 + M^2)^2 + M^3(M^2 + M^2/2) + M^2MM^2.$$

This constant is integrable on $[0, 1]$. Every approximating integrand is continuous and measurable. At almost every point the two primitives and curvature converge, while the original probe and its derivatives remain fixed. Polynomial continuity therefore gives convergence of the entire density, and dominated convergence gives the stated action limit. No derivative convergence of the curvature has been used. $\square$

Lemma 21.4. *Fix the smooth negative area probe r from Theorem 20.11, and set*

$$\begin{aligned} K_t(x) &= (1 - t)K_*(x) + 2t, \\ \kappa_t &= (1 - t)L_* + 2t, \quad \Lambda_t = (1 - t)L_*C_{CK} + 2t. \end{aligned} \tag{79}$$

There is $0 < t < 1$ such that $A_{K_t}(r) < 0$. For this same t , the curvature is continuous on $\mathbb{R}$, has both exact normalized moments, is symmetric on $[0, 1]$ and has the same two hemisphere order laws. Its box and its ratio satisfy

$$\begin{aligned} 0 < \kappa_t < 2 < \Lambda_t, \quad \kappa_t \leq K_t(x) \leq \Lambda_t \quad (x \in \mathbb{R}), \\ C_{CK} - \frac{\Lambda_t}{\kappa_t} &= \frac{2t(C_{CK} - 1)}{\kappa_t} > 0. \end{aligned} \tag{80}$$

The probe r , including its derivatives and endpoint data, is unchanged.

Proof. Both moments are affine in curvature; the constant curvature two has moments two and one. A nonnegative coefficient $1 - t$ preserves the physical reflection and order laws and the original global box. The strict bounds follow from $L_* < 2 < L_*C_{CK}$. Multiplying the ratio

difference by the positive denominator κ_t cancels the two copies of $(1-t)L_*C_{CK}$, giving exactly (80).

For the actual primitives, linearity of the integral gives

$$f_{K_t} = (1-t)f_{K_*} + tf_2, \quad f_{1,K_t} = (1-t)f_{1,K_*} + tf_{1,2}.$$

Substitute these into the complete density with the fixed r, r', r'' . The resulting function of (t, x) is continuous, so its integral over the compact physical interval is continuous in t . Its value at zero is strictly negative. A sufficiently small positive $t < 1$ therefore retains negative action. The ratio deficit is obtained before any smoothing. $\square$

Theorem 21.5. *There are $0 < t < 1$, functions $K, r : \mathbb{R} \rightarrow \mathbb{R}$ smooth on $\mathbb{R}$, produced sphere data D , and globally smooth maps $X, R : \mathbb{R} \rightarrow \mathbb{R}$ with the following properties. The same area probe r equals one near both endpoints, has $r(0) = 1$, $r'(0) = 0$, and is nonconstant on $[0, 1]$. On $[0, 1]$, the recovered K is symmetric with the two hemisphere order laws and satisfies*

$$\kappa_t \leq K \leq \Lambda_t, \quad \int_0^1 K = 2, \quad \int_0^1 xK(x) dx = 1, \quad A_K(r) < 0,$$

where the identical contracted box obeys (80). All endpoint, interval-mapping, inverse and actual probe-jet laws of Theorem 20.6 hold for these same K, r, X, D, R . In particular $R = r \circ X$ globally, and for every $x \in [0, 1]$,

$$D.a(V_K(x)) = \frac{1}{K(x)}, \quad R(V_K(x)) = r(x).$$

For every sphere point p one has $K_{g_D}(p) = K(X(z(p)))$. The original smooth one-form $h = M_D(R)$ is nonzero and satisfies

$$g_D \in \mathcal{G}_{\text{rot}}, \quad \kappa_t \leq K_{g_D} \leq \Lambda_t, \quad \frac{\Lambda_t}{\kappa_t} < C_{CK}, \quad (81)$$

$$Q_{g_D}(h) = 2\pi A_K(r) < 0.$$

Proof. Choose the fixed probe and a positive contraction parameter from Lemma 21.4. Apply Theorem 21.2 with target mean two and exactly the box $[\kappa_t, \Lambda_t]$. Reindex by its almost-everywhere convergent subsequence. The strictly increasing index tends to infinity, so the reindexed sequence still converges in L^1 and retains every moment, shape and box condition. Lemma 21.3, applied to the same r , gives convergence of the complete actions to the already negative contracted action. Choose an actual negative member and call it K .

This K is smooth on $\mathbb{R}$, positive on $[0, 1]$ and has both exact moments. Theorem 20.6 therefore constructs the stated X, D, R and retains all of their equations for the same K, r . The original action identity and $2\pi > 0$ give strict negativity. Since the zero form has zero action, the original form is nonzero. The physical range of X places $X(z(p))$ in $[0, 1]$ for every sphere point, so the global curvature composition transports the identical box. The identity diffeomorphism places g_D in the represented class. The strict ratio is the one already proved before recovery. $\square$

Lemma 21.6. *Let (Σ, g) be a compact smooth Riemannian surface without boundary whose actual Gauss curvature is the constant $\kappa \in \mathbb{R}$. For every smooth one-form h ,*

$$Q_g(h) = \int_{\Sigma} (|\Delta^{\nabla} h + \kappa h|_g^2 + \kappa |\nabla h|_g^2 + 2\kappa |A_g h|_g^2) d\mu_g. \quad (82)$$

Every term is integrable. If $\kappa \geq 0$, then $Q_g(h) \geq 0$ for every smooth one-form, with no conformal-Killing or invariance condition.

Proof. The actual smooth tensors and their inner products have integrable norms on the compact surface. Canonical Green integration gives

$$\int_{\Sigma} \langle h, \Delta^{\nabla} h \rangle_g d\mu_g = - \int_{\Sigma} |\nabla h|_g^2 d\mu_g.$$

Expanding the first square in (82), its cross term contributes $-2\kappa \int |\nabla h|^2$. Combining the displayed positive derivative term leaves exactly the $-\kappa |\nabla h|^2$ term in the original four-term definition (27). The other two terms are unchanged. This proves the identity; when $\kappa \geq 0$ its right-hand density is nonnegative pointwise. $\square$

Define $\mathcal{S}_{\text{rot}}$ to consist of all real $C \geq 1$ such that $Q_g(h) \geq 0$ for every $g \in \mathcal{G}_{\text{rot}}$, every positive actual curvature box $\kappa > 0$, $\kappa \leq K_g \leq \Lambda$ with $\Lambda/\kappa \leq C$, and every original smooth one-form h . Put $C_{\text{rot}} = \sup \mathcal{S}_{\text{rot}}$.

Theorem 21.7. *The unrestricted safe-cap set contains one, is downward closed among caps at least one, and is bounded above. For a contraction parameter $0 < t < 1$ supplied by the smooth negative witness,*

$$1 \leq C_{\text{rot}} \leq \frac{\Lambda_t}{\kappa_t} < C_{\text{CK}} = \sup \mathcal{S}_{\text{CK}}. \quad (83)$$

Thus the unrestricted rotational threshold is strictly smaller than the actual universal conformal-Killing threshold.

Proof. A positive curvature box with ratio at most one satisfies $\Lambda \leq \kappa$. Its pointwise bounds therefore force $K_g = \kappa$. Lemma 21.6 proves safety for every smooth form, so $1 \in \mathcal{S}_{\text{rot}}$. Downward closure follows directly from the universal box quantifier.

Use the actual smooth witness of Theorem 21.5 and put $b = \Lambda_t/\kappa_t$. If a safe cap C were at least b , its universal safety assertion would apply to that same metric, positive box and nonzero original form, contradicting its negative action. Consequently every safe cap is strictly below b . The safe set is nonempty and bounded above; the real supremum laws give $1 \leq C_{\text{rot}} \leq b$. Its strict separation from the actual CK supremum now follows from (80) and Theorem 19.7. $\square$

The exact value of C_{rot} and smooth attainment at its unrestricted endpoint are not determined here.

References

- [1] D. Jerison, *Fourier Series, Part 1*, MIT 18.103 Fourier Analysis, Fall 2013, Corollary 2, p. 5. MIT OpenCourseWare. Lecture notes.